

cut-condition.cc

Lambda cut study from  $p + \pi^-$

run02447

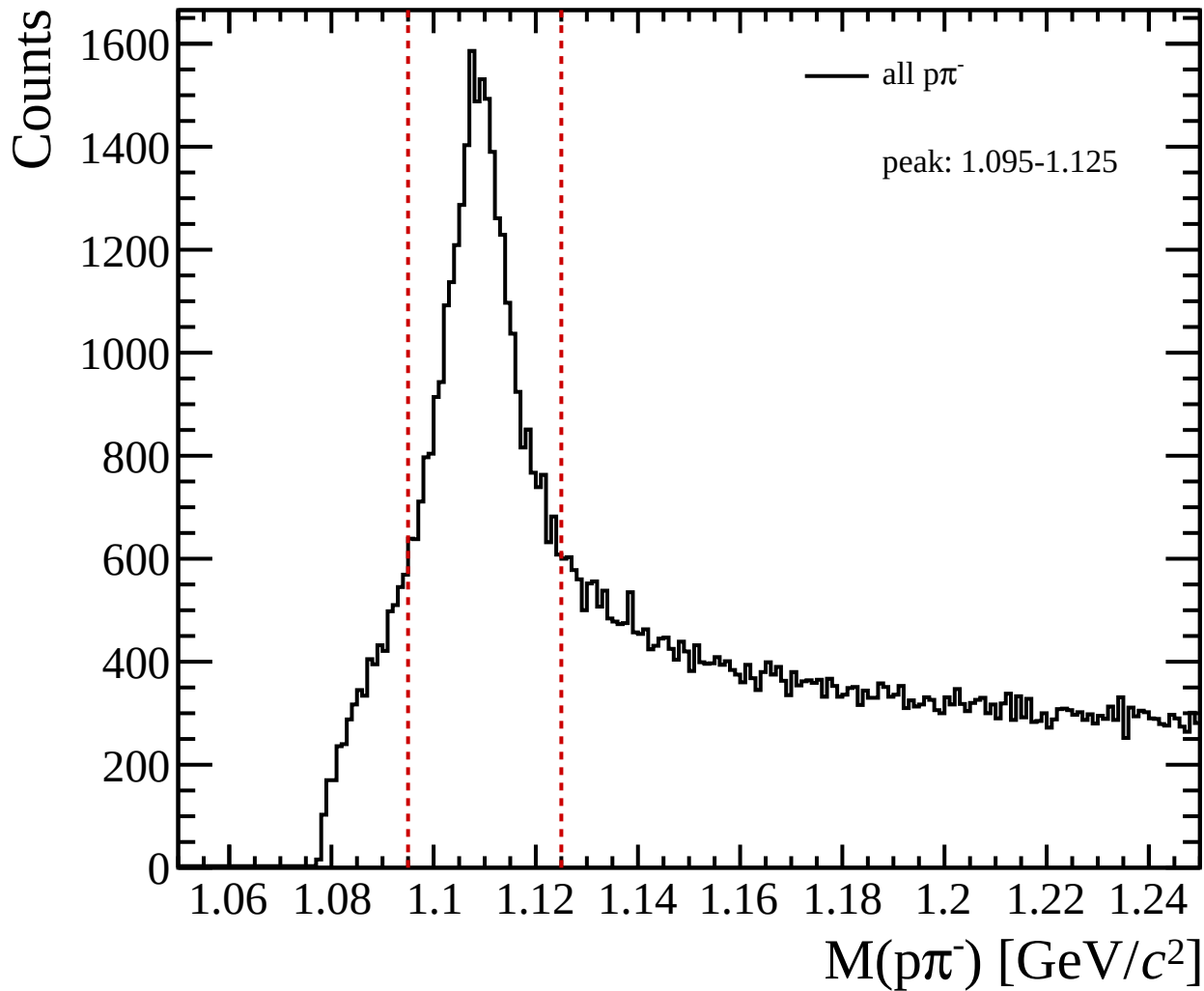
comparison: all  $p\pi^-$ , Lambda peak, and tight quality cut

peak window:  $1.095 < M(p\pi^-) < 1.125 \text{ GeV}/c^2$

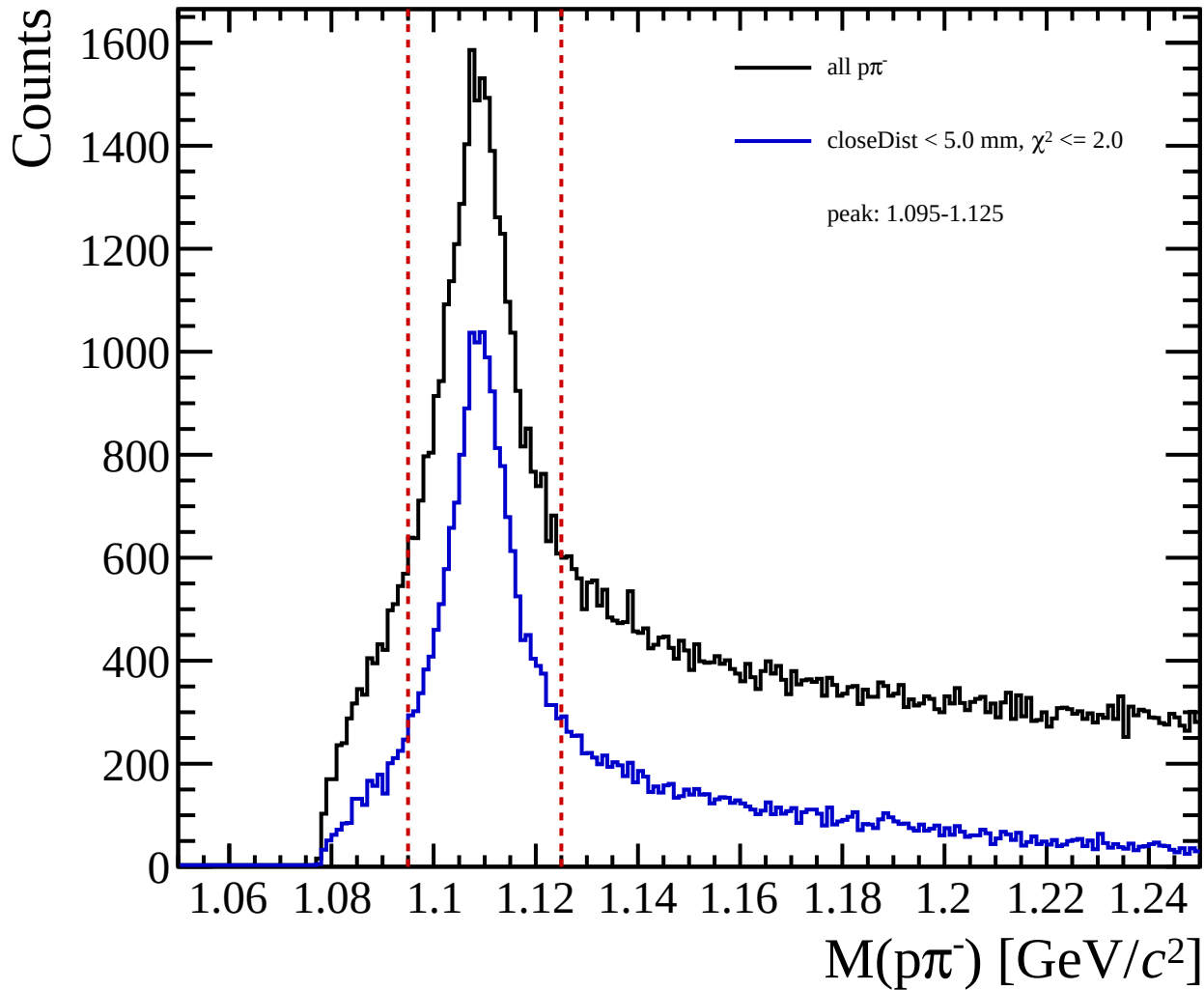
tight cut:  $\text{closeDist} < 5.0 \text{ mm}$ ,  $\chi_p^2 \leq 2.0$ ,  $\chi_\pi^2 \leq 2.0$

Generated at: 2026-07-06 18:13:38

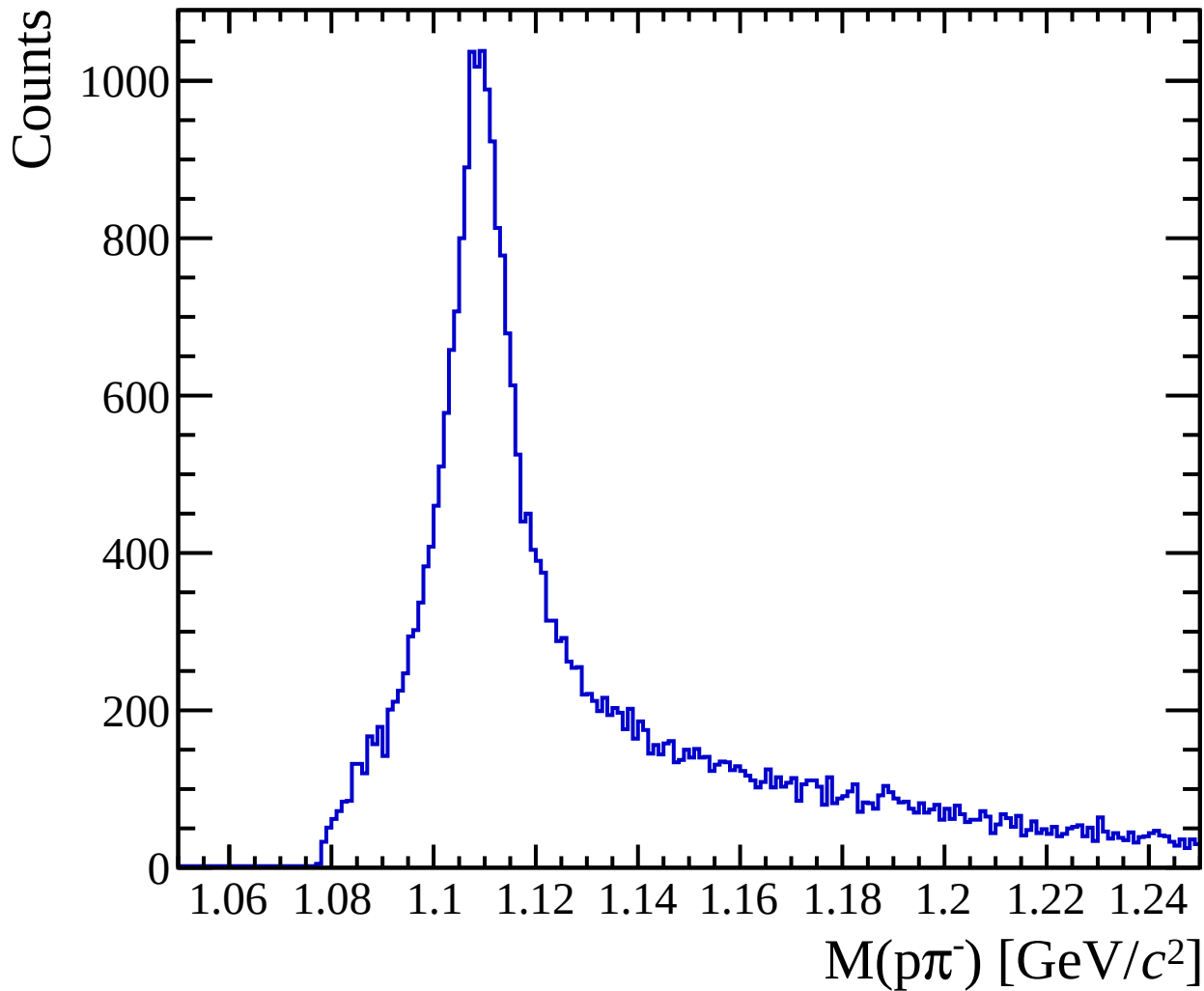
# $p\pi^-$ invariant mass



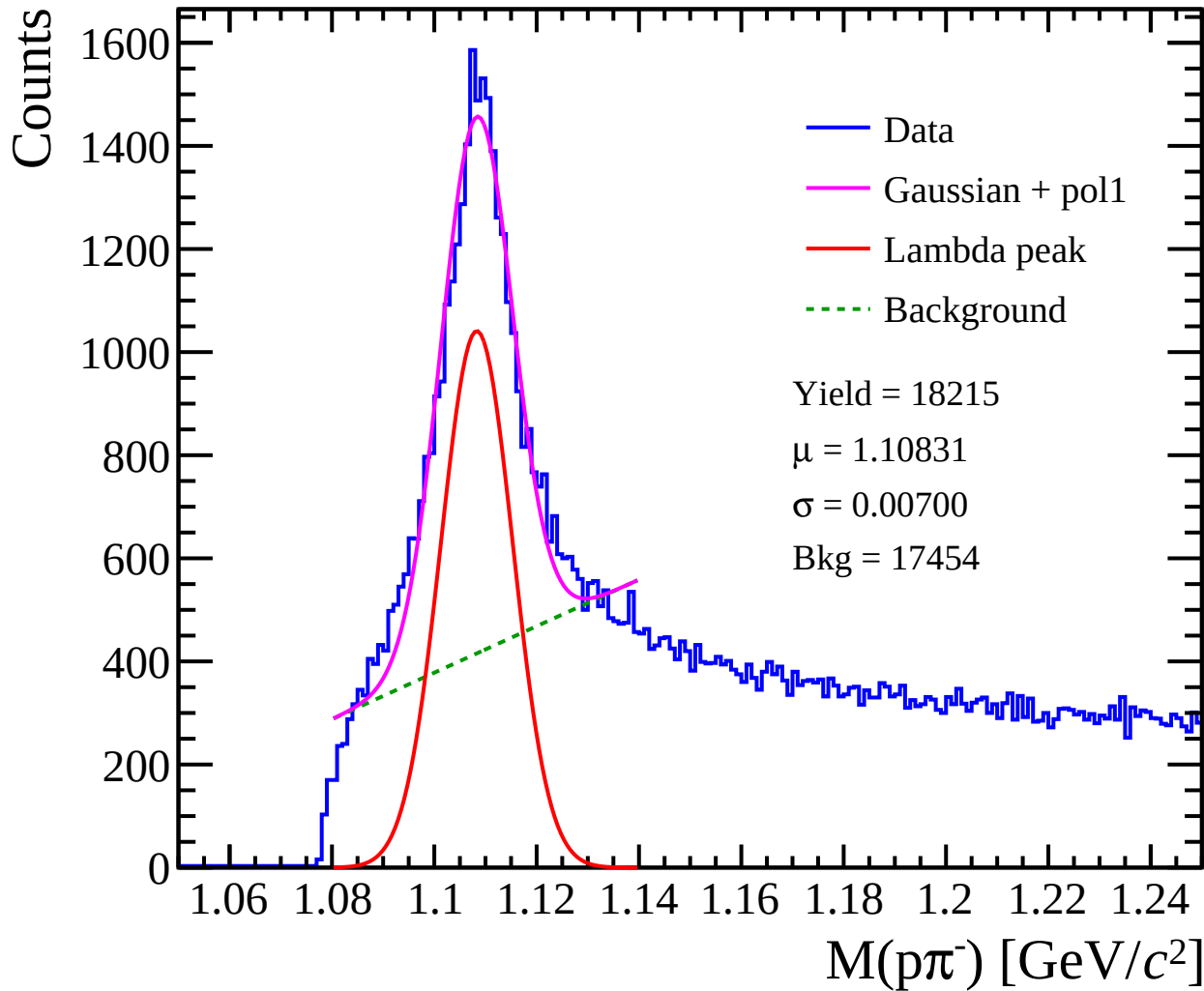
# $p\pi^-$ invariant mass



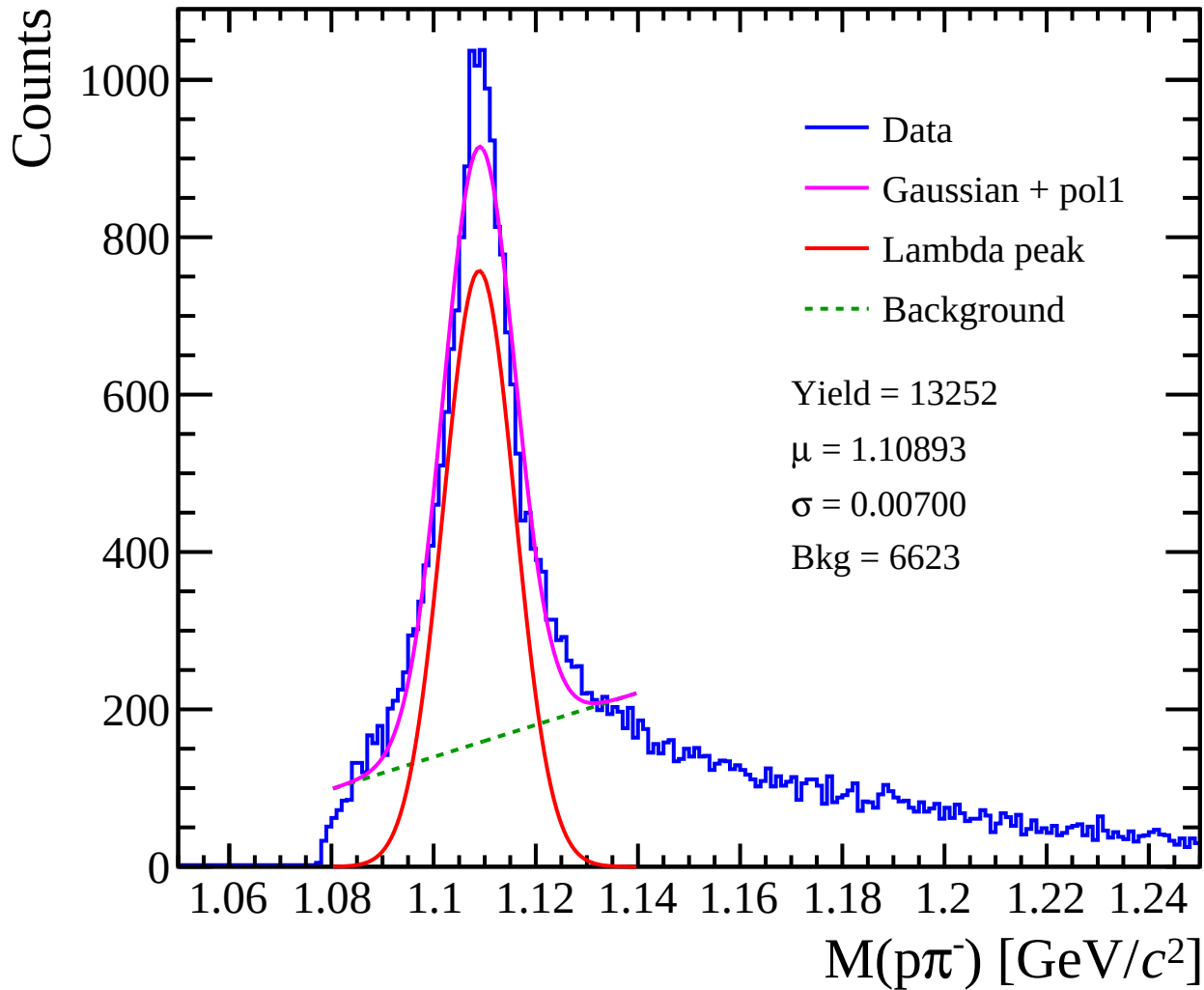
$p\pi^-$  invariant mass, tight cut



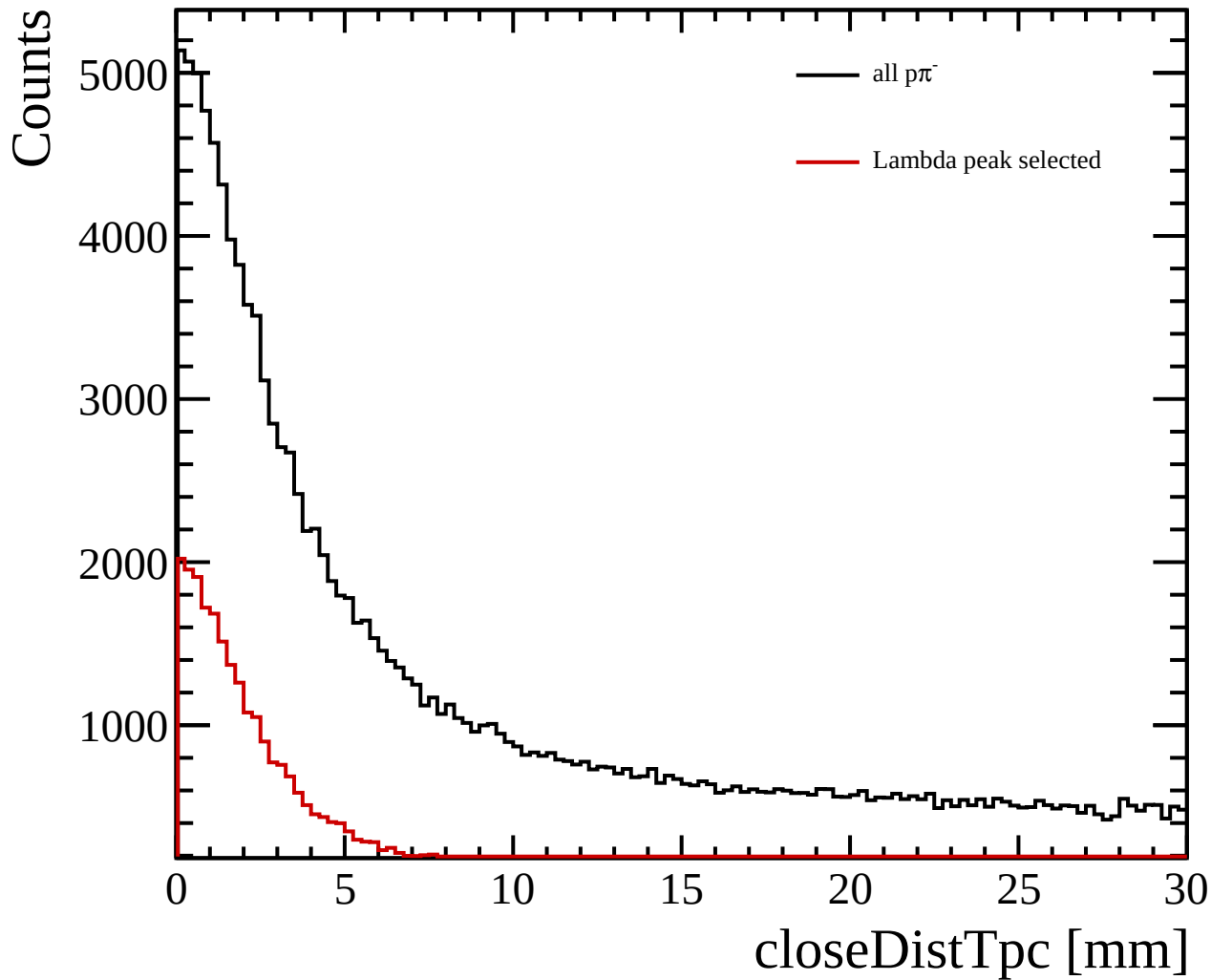
# $p\pi^-$ invariant mass



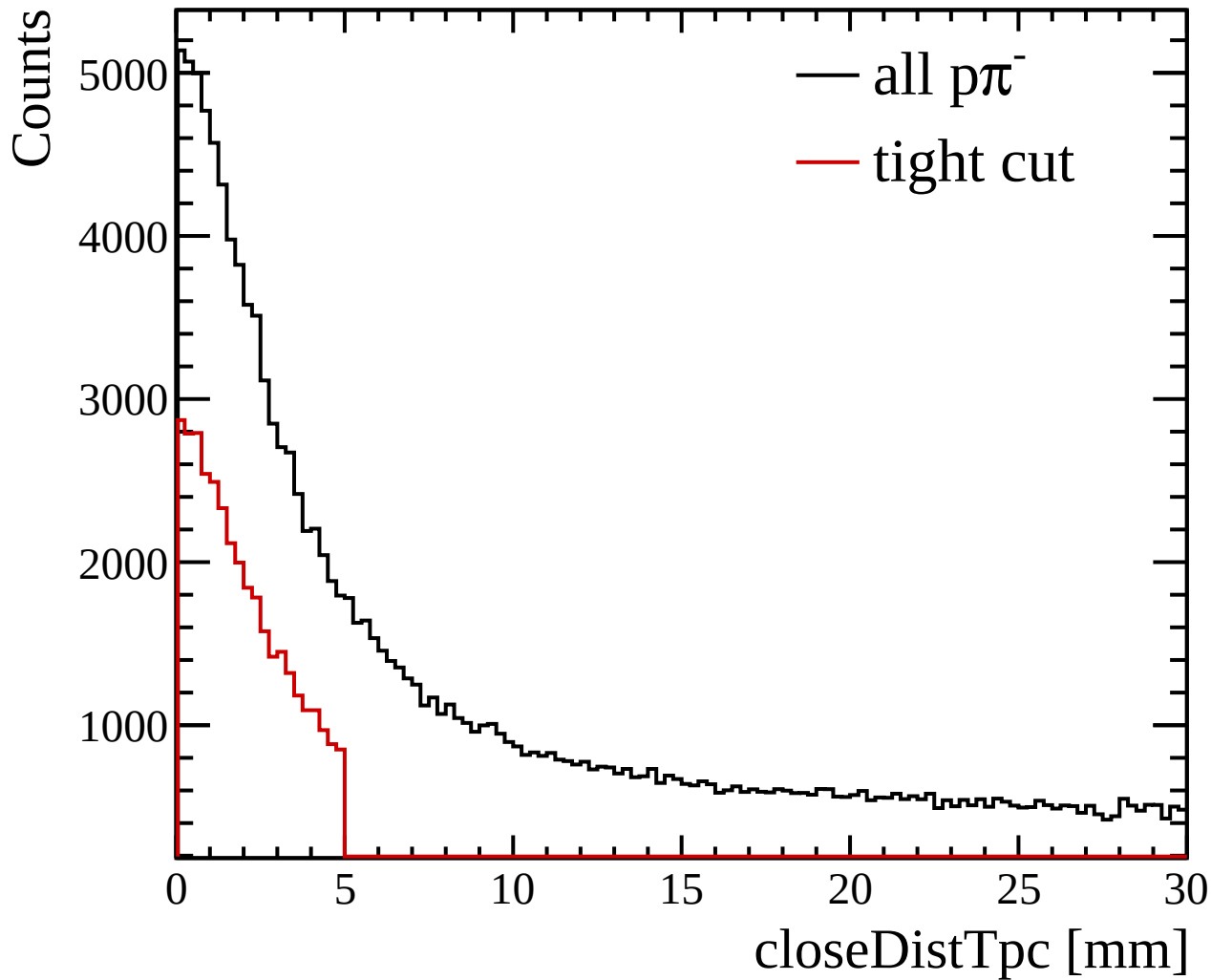
# $p\pi^-$ invariant mass, tight cut



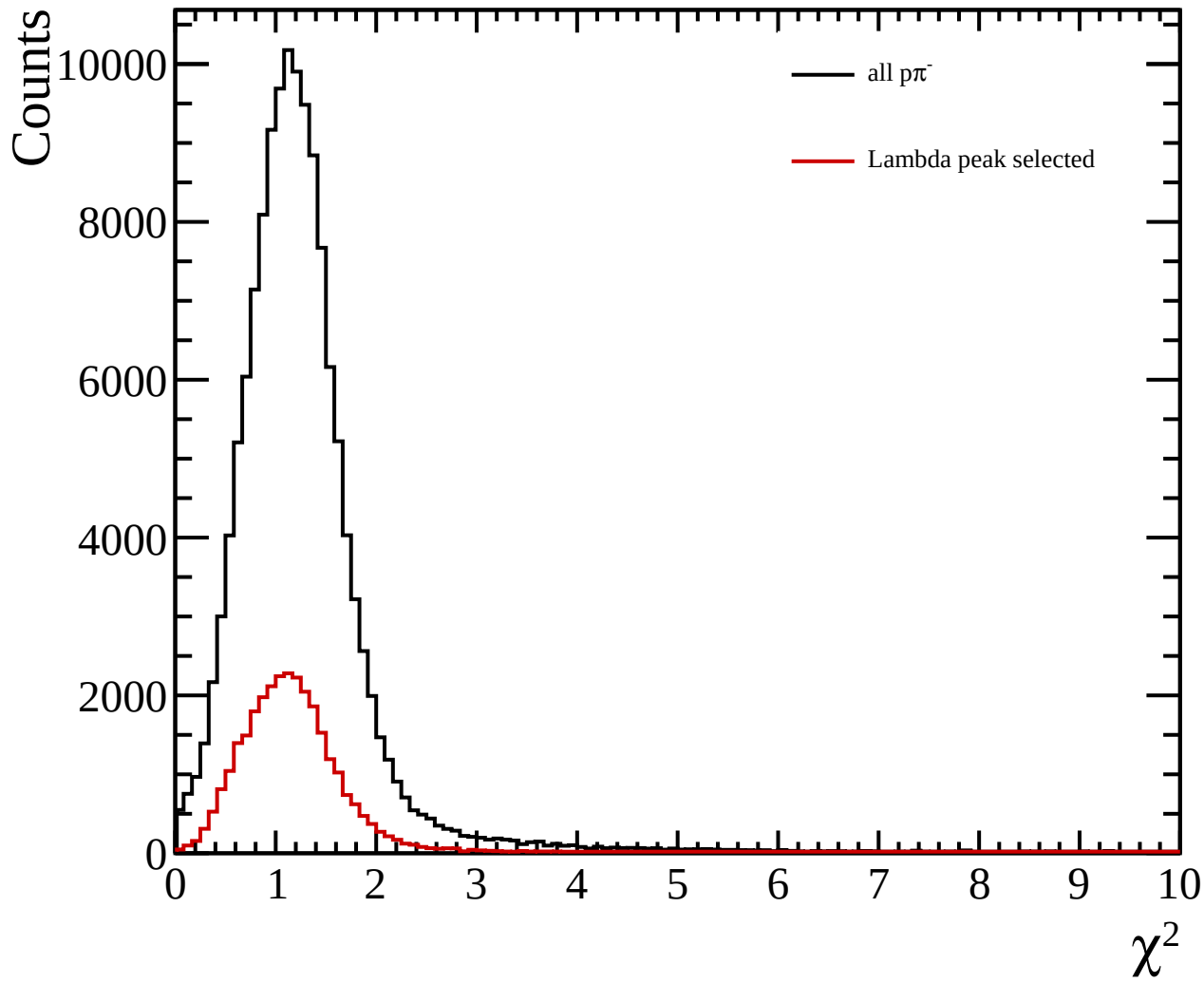
# $p\pi^-$ close distance



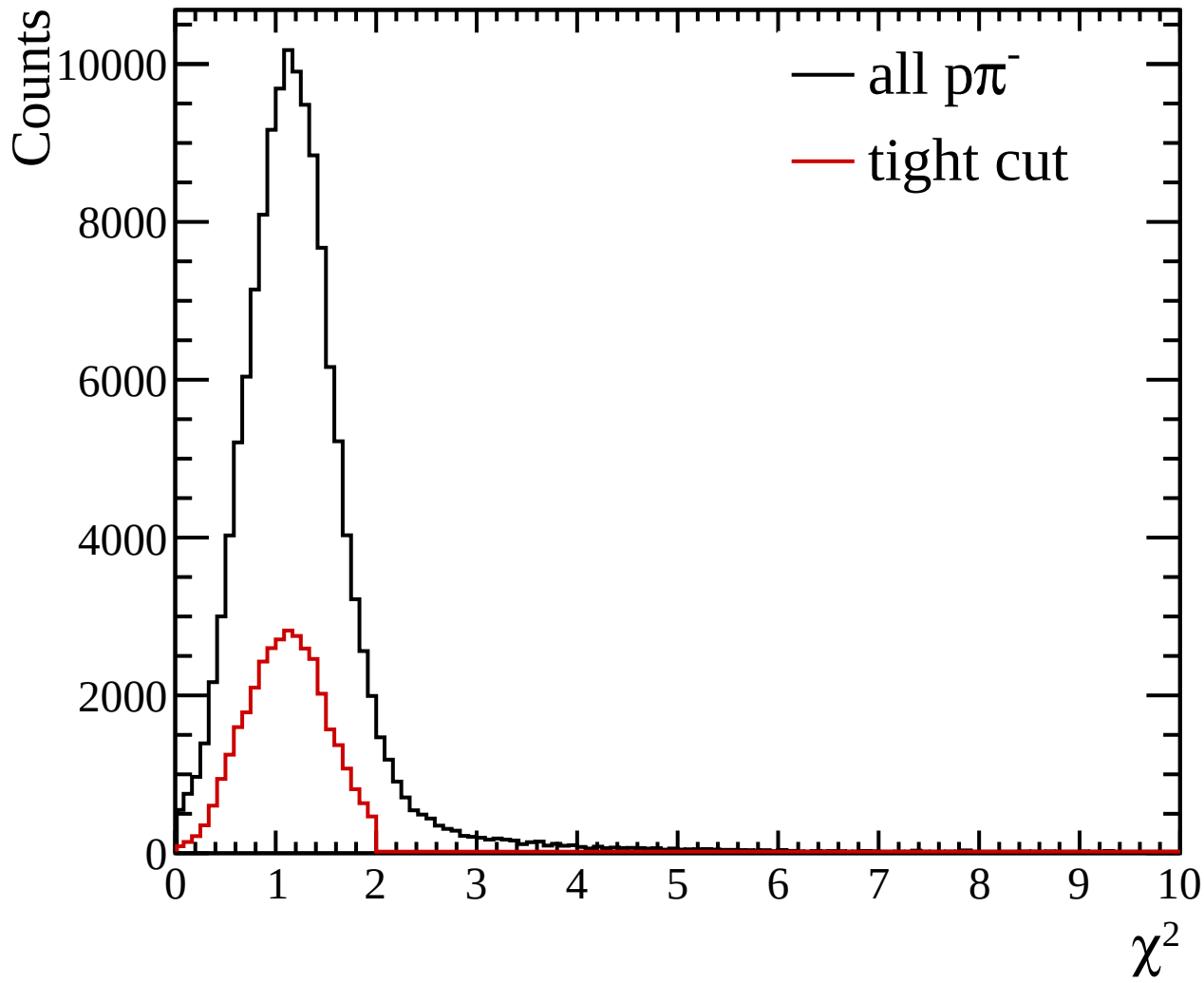
# $p\pi^-$ close distance



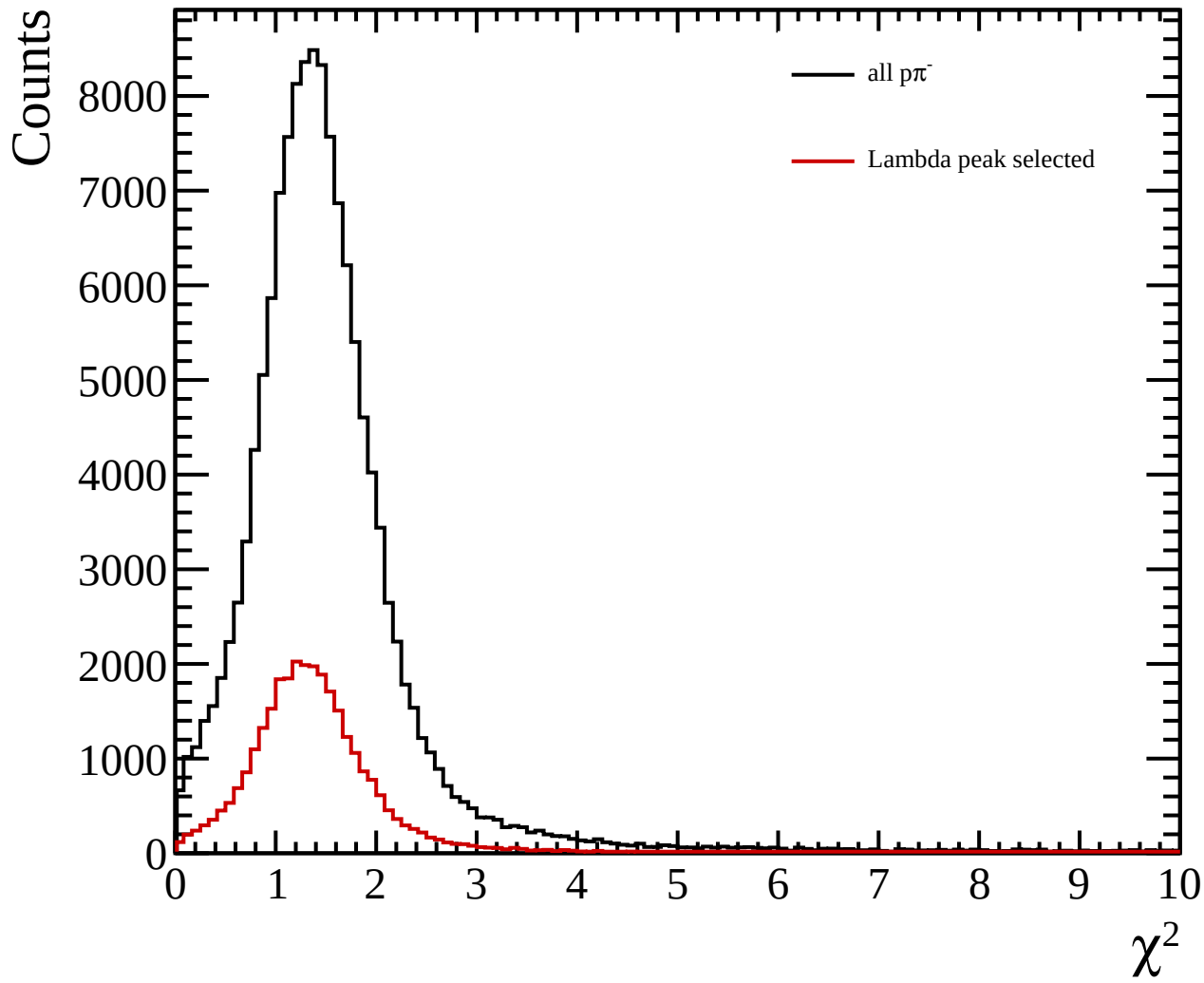
# proton track $\chi^2$



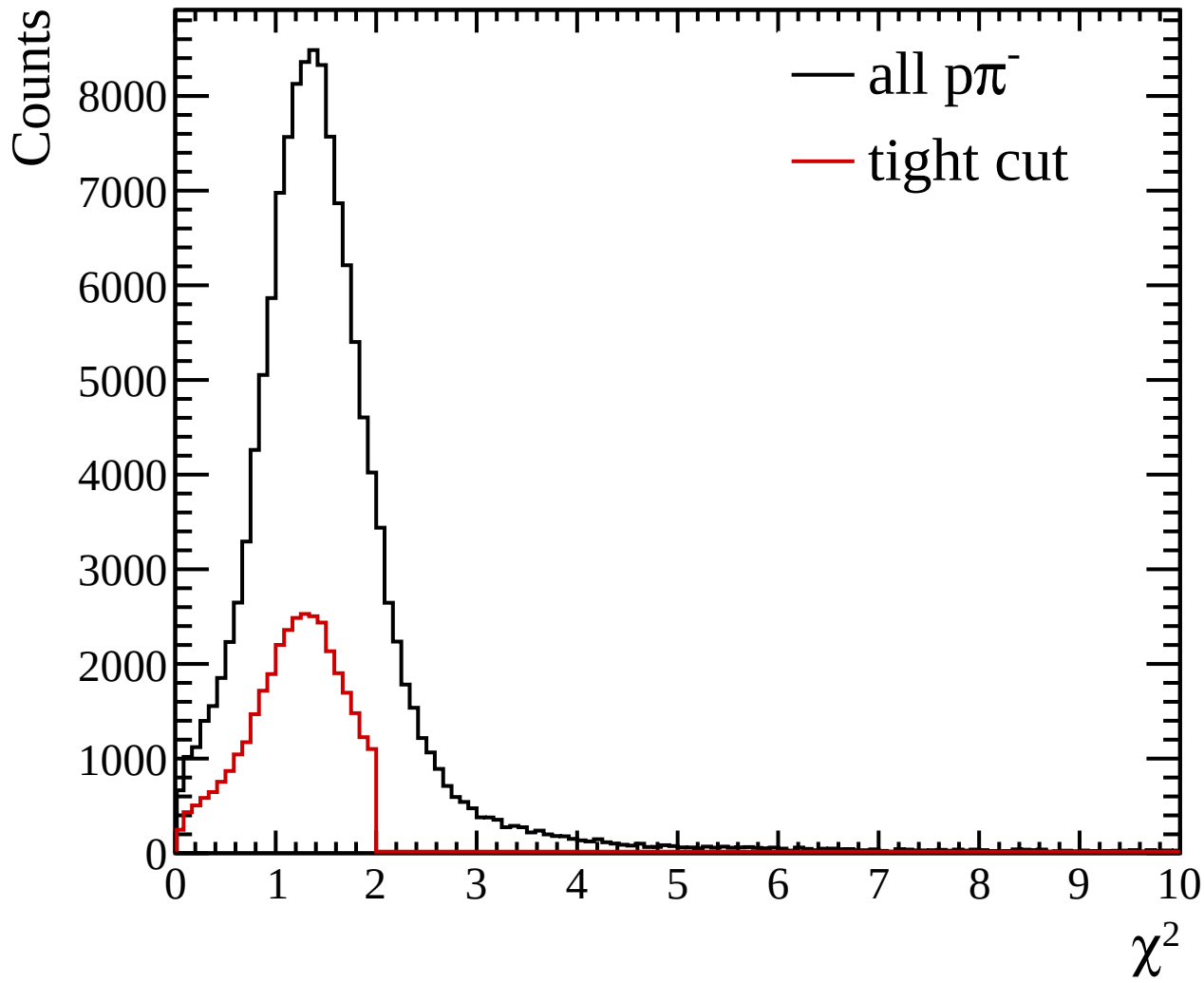
proton track  $\chi^2$



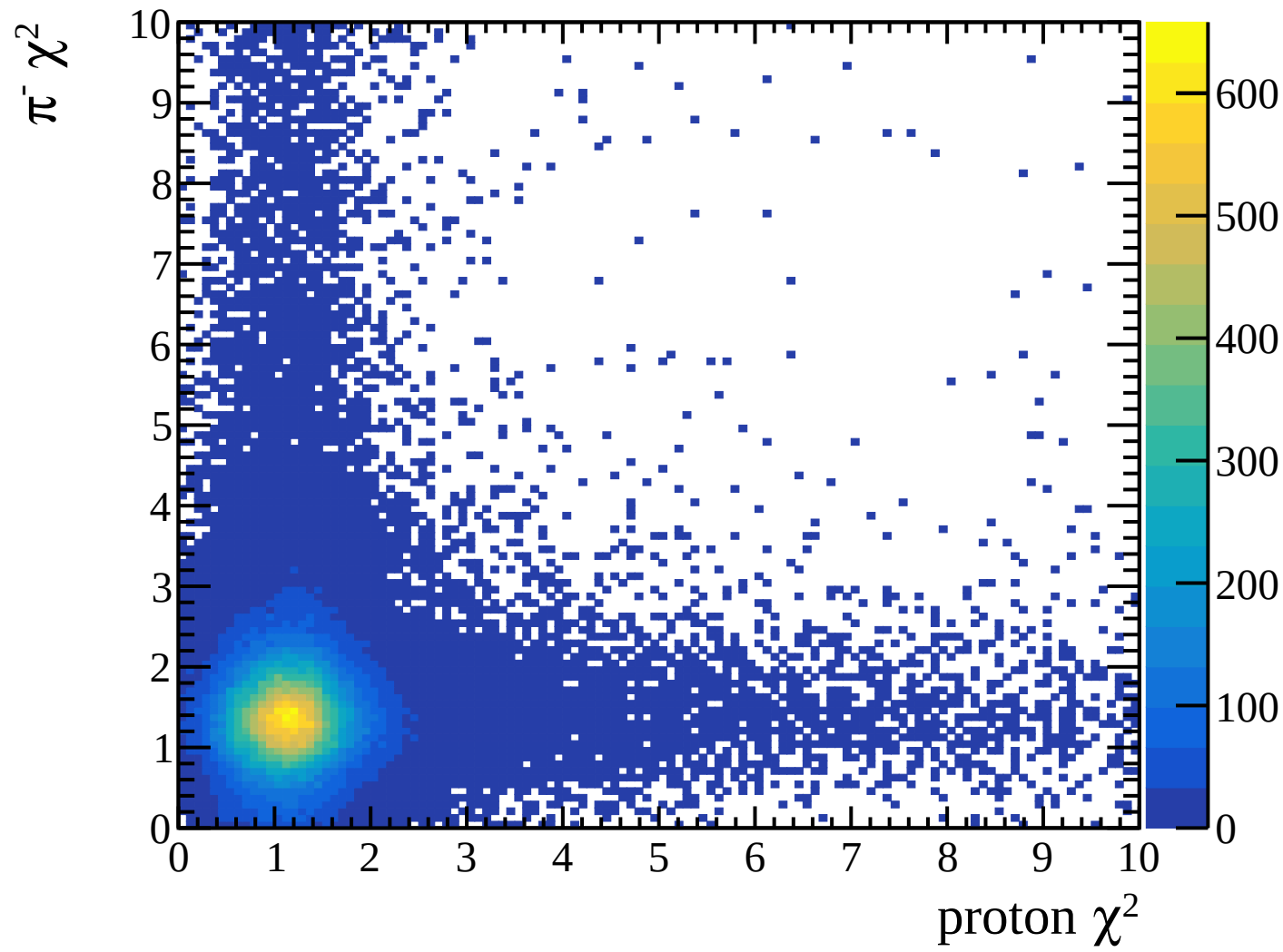
$\pi^-$  track  $\chi^2$



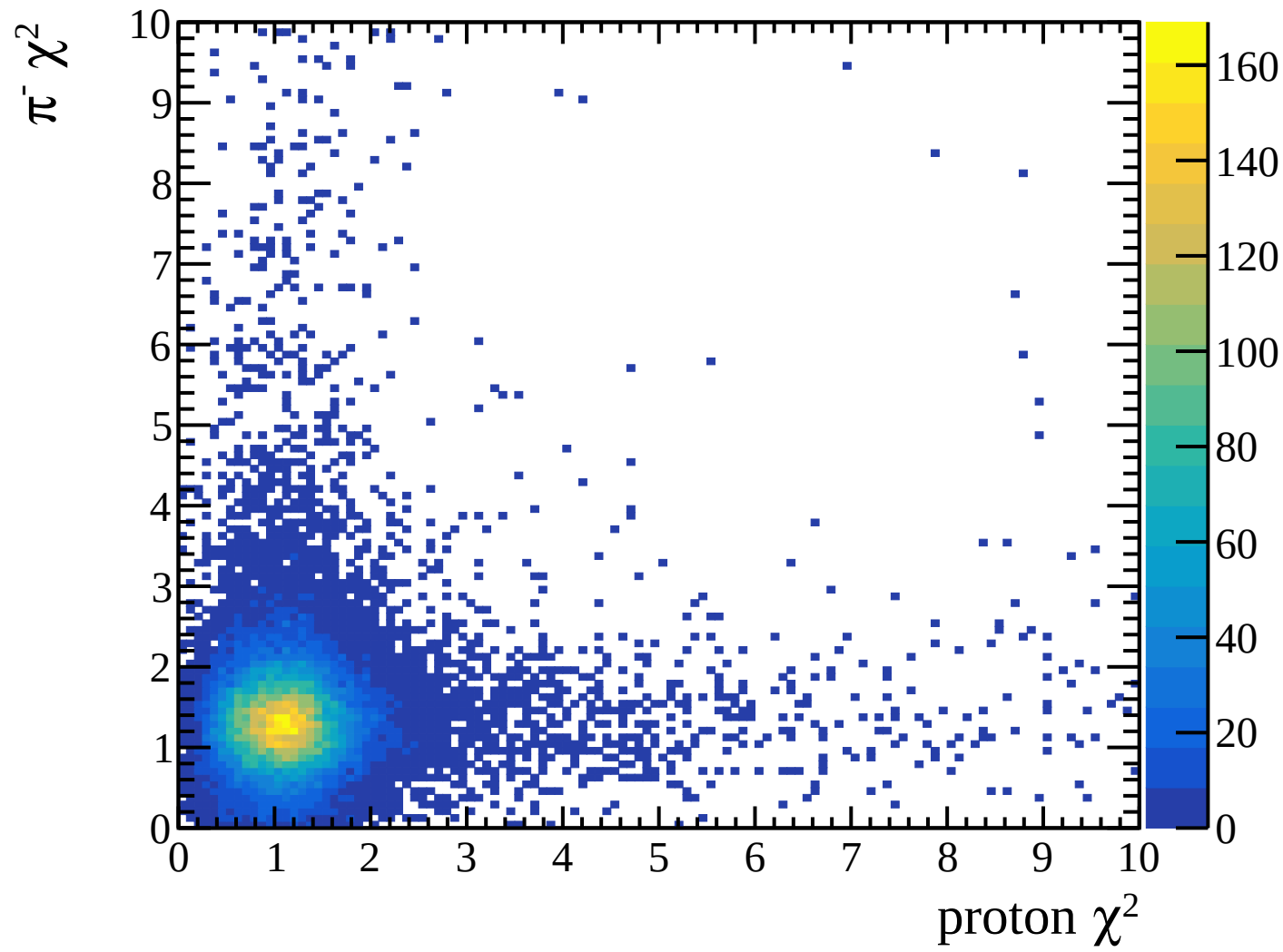
$\pi^-$  track  $\chi^2$



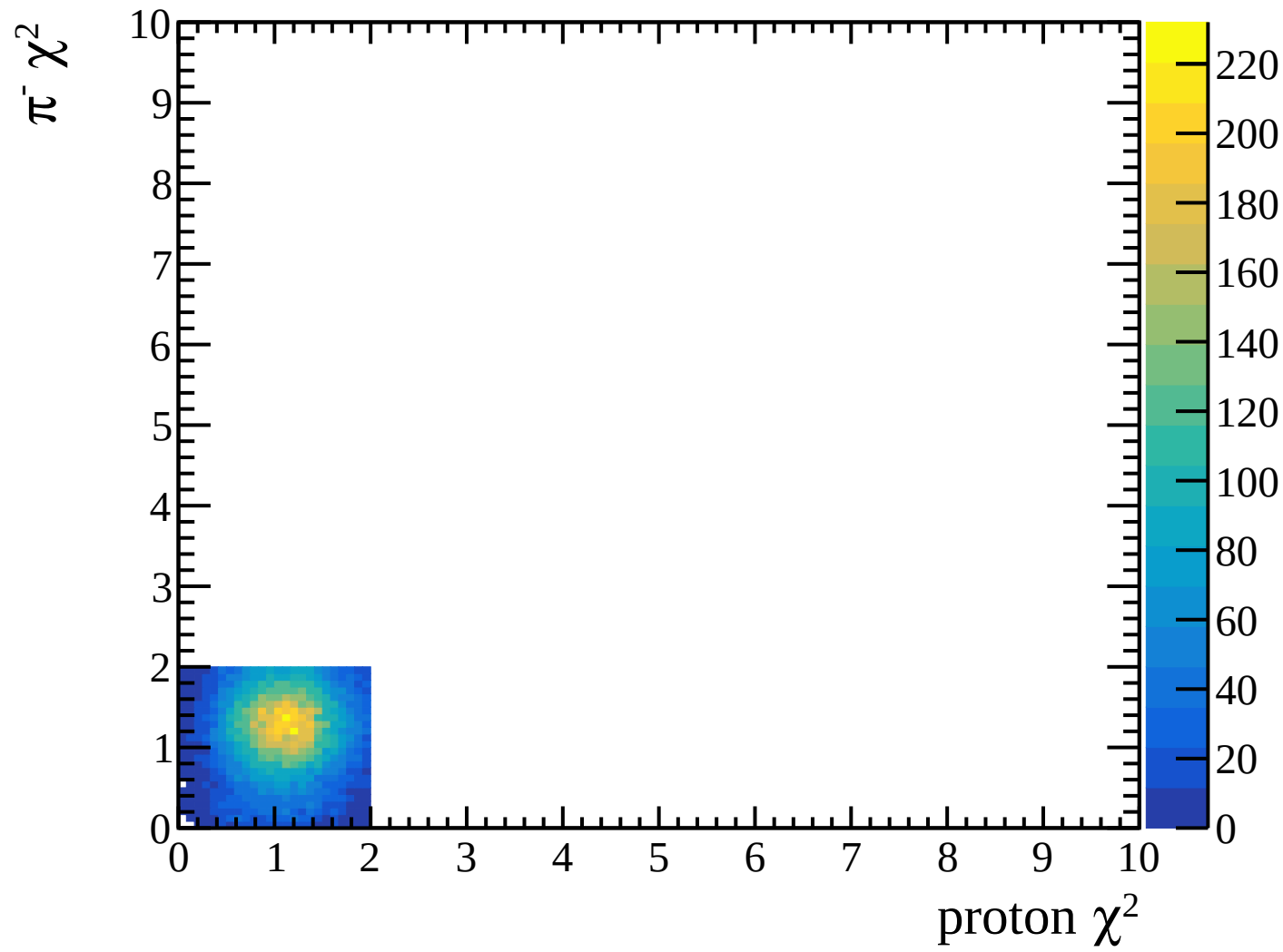
track  $\chi^2$ , all  $p\pi^-$



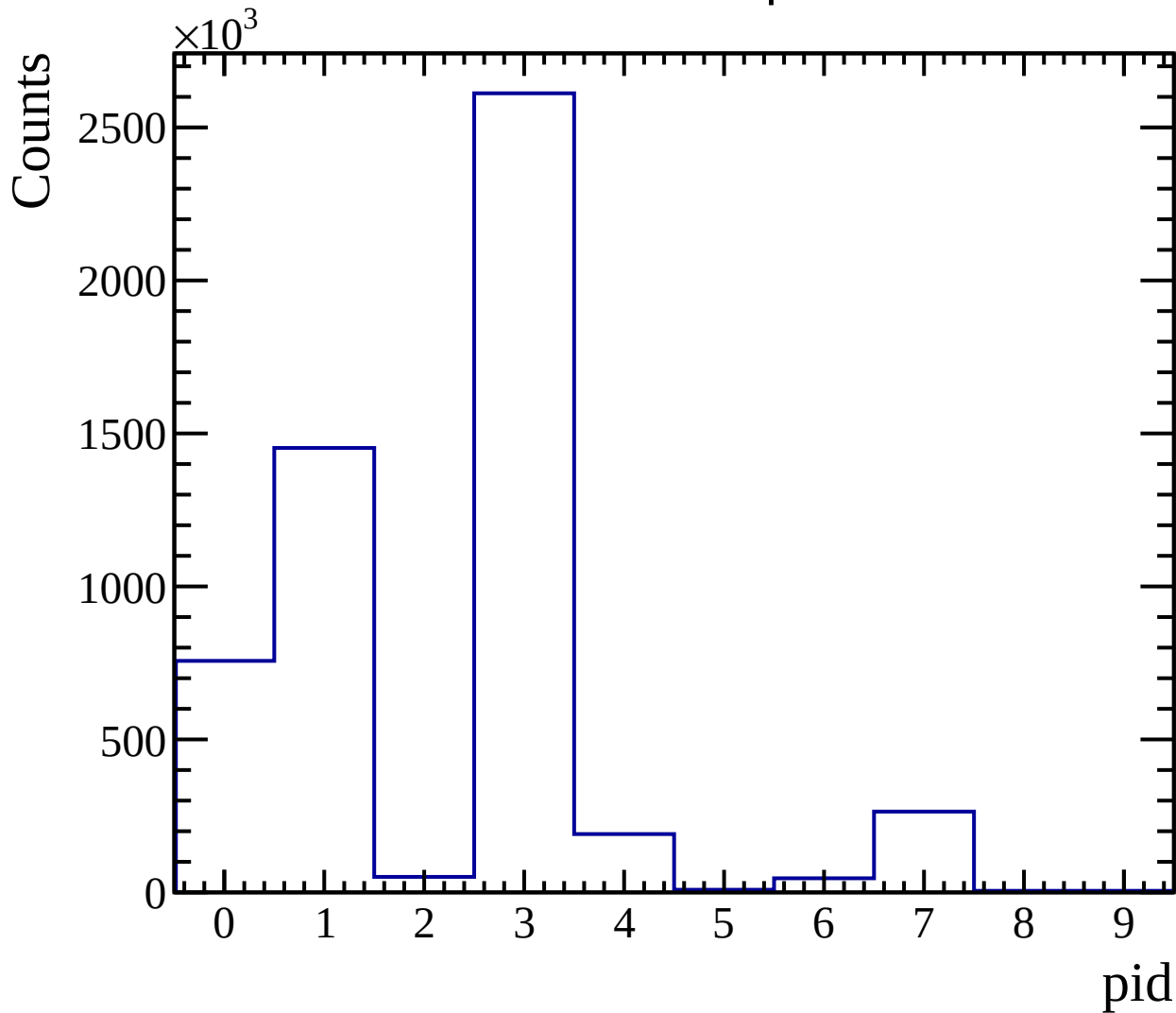
track  $\chi^2$ , Lambda peak



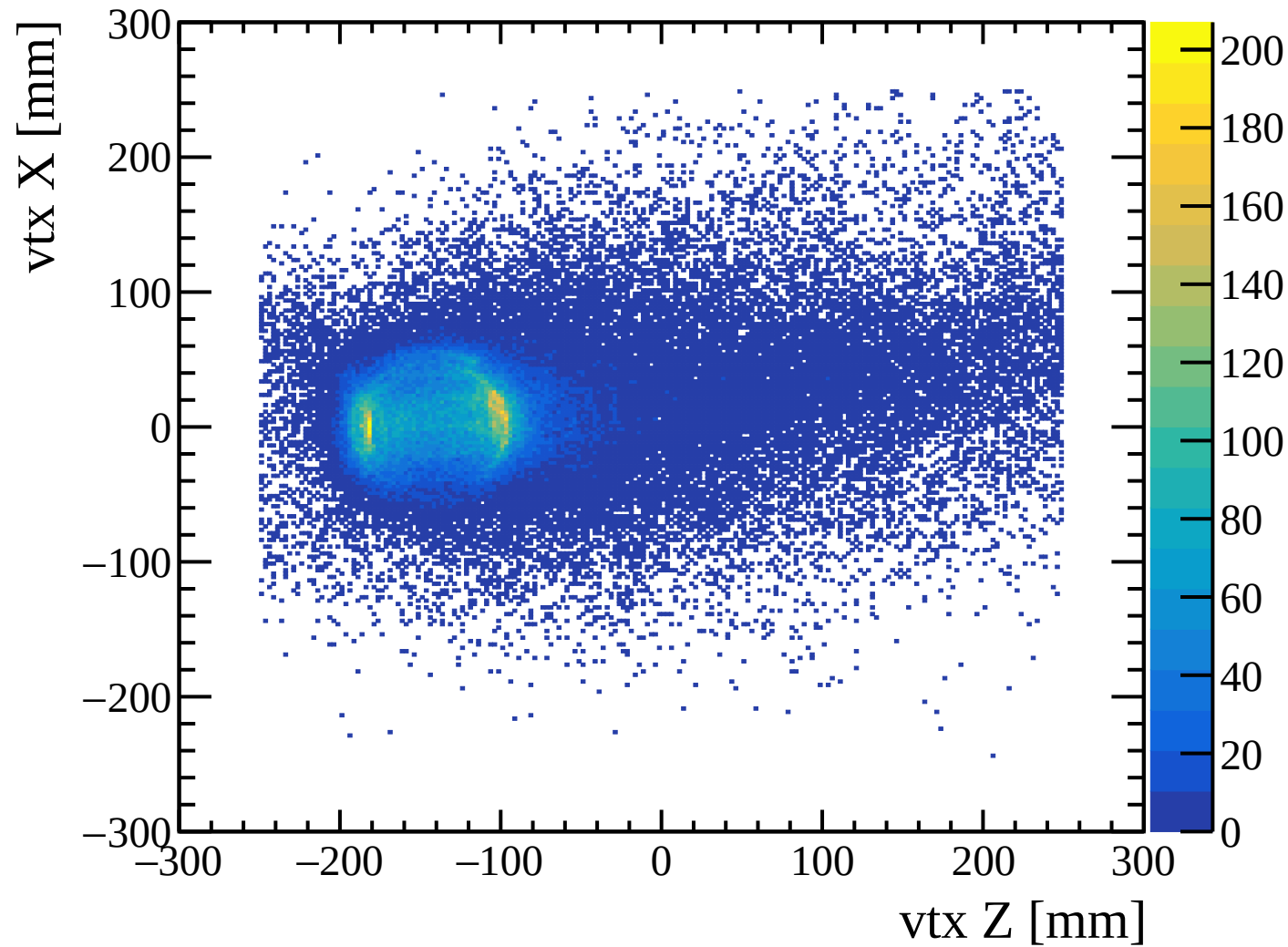
track  $\chi^2$ , tight cut



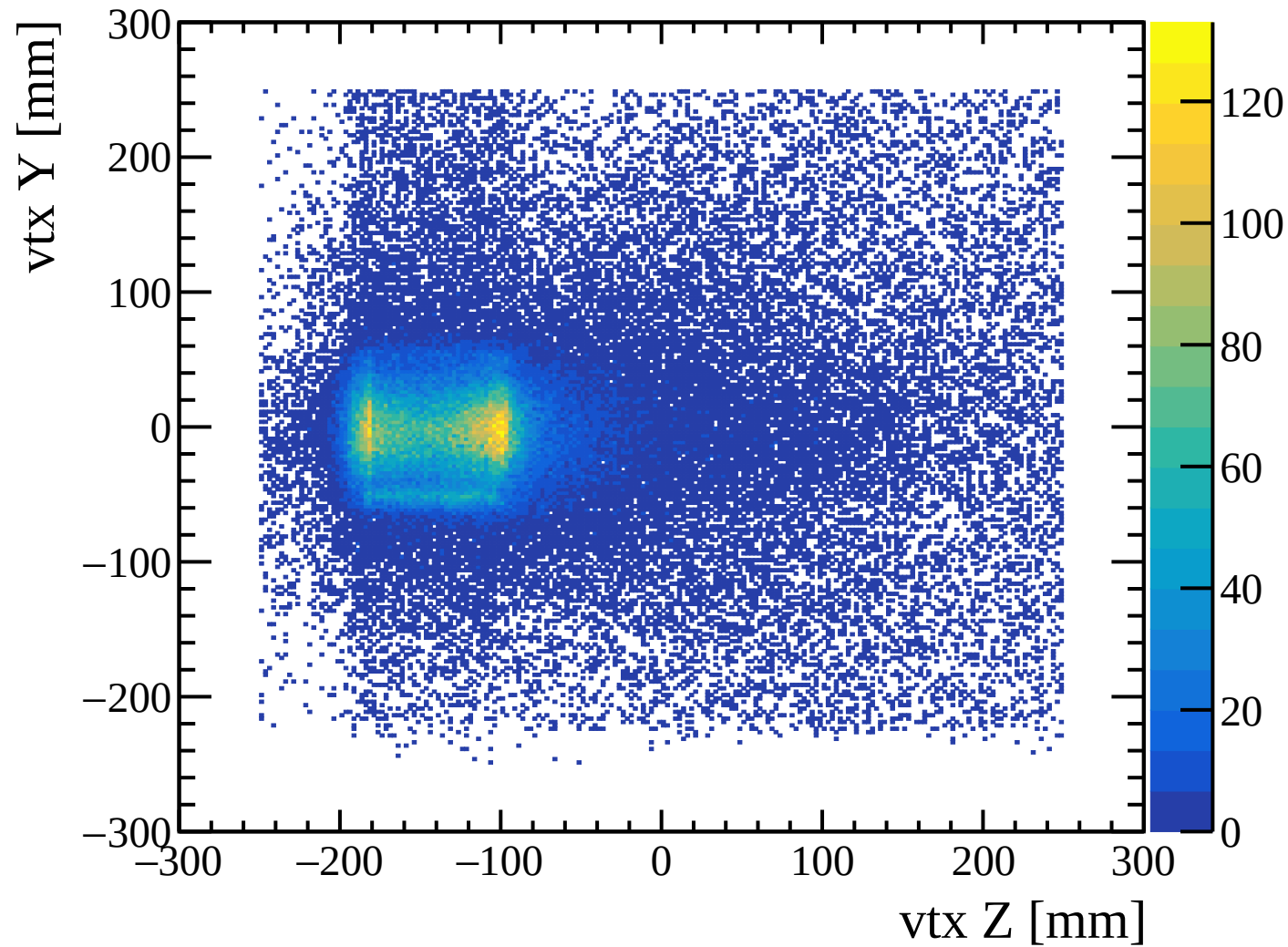
track PID bit pattern



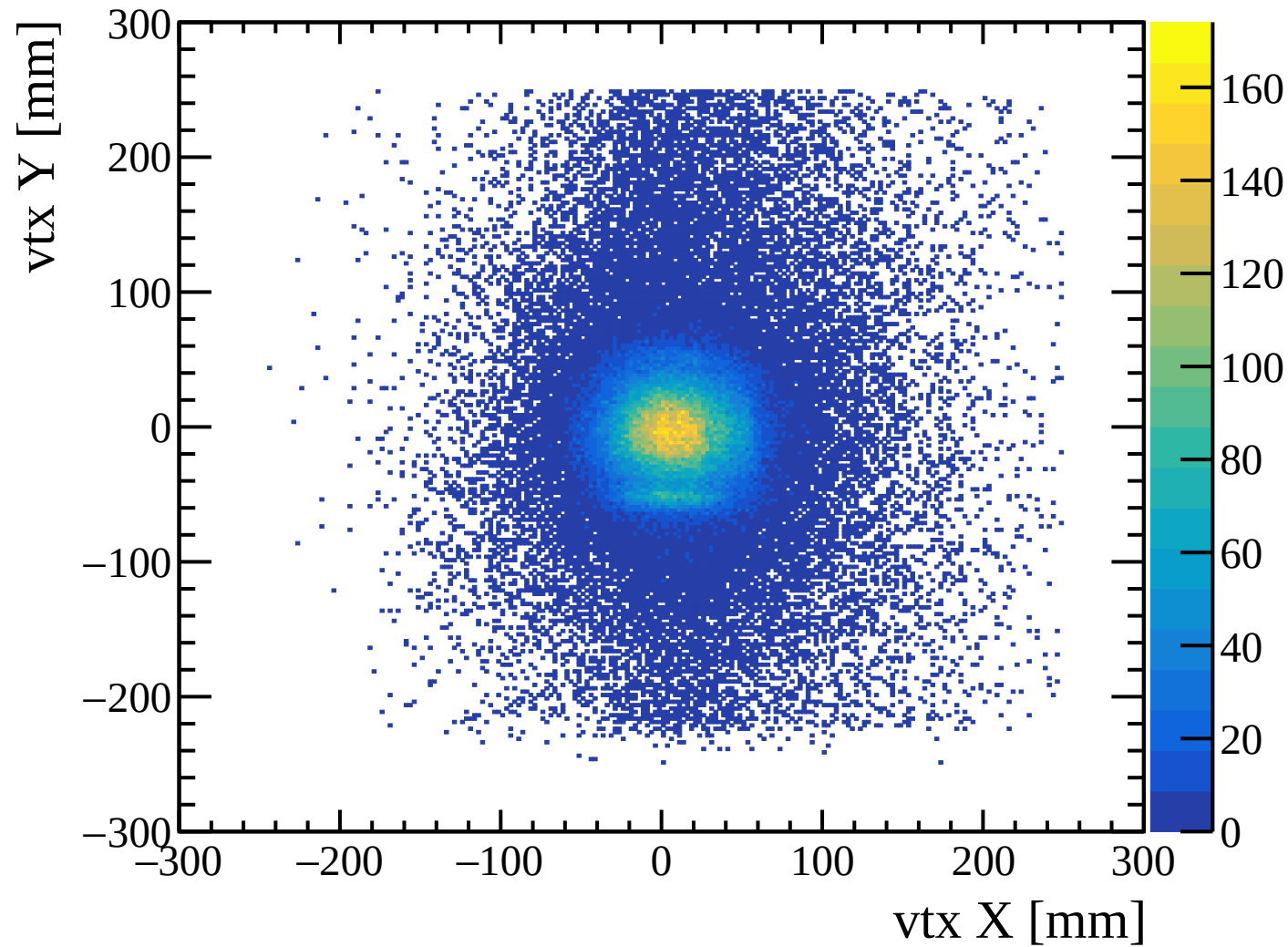
vertex, all  $p\pi^-$



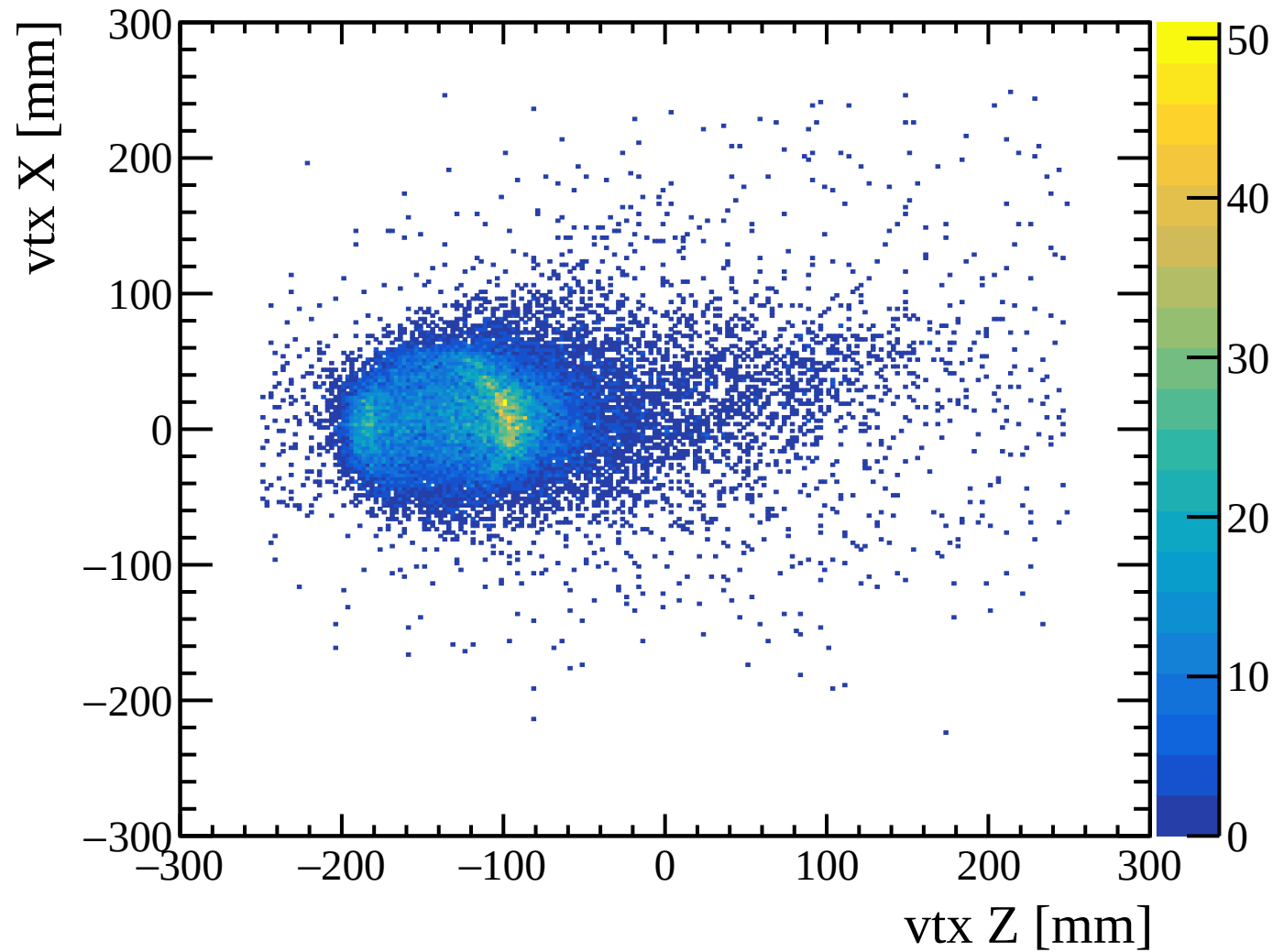
vertex, all  $p\pi^-$



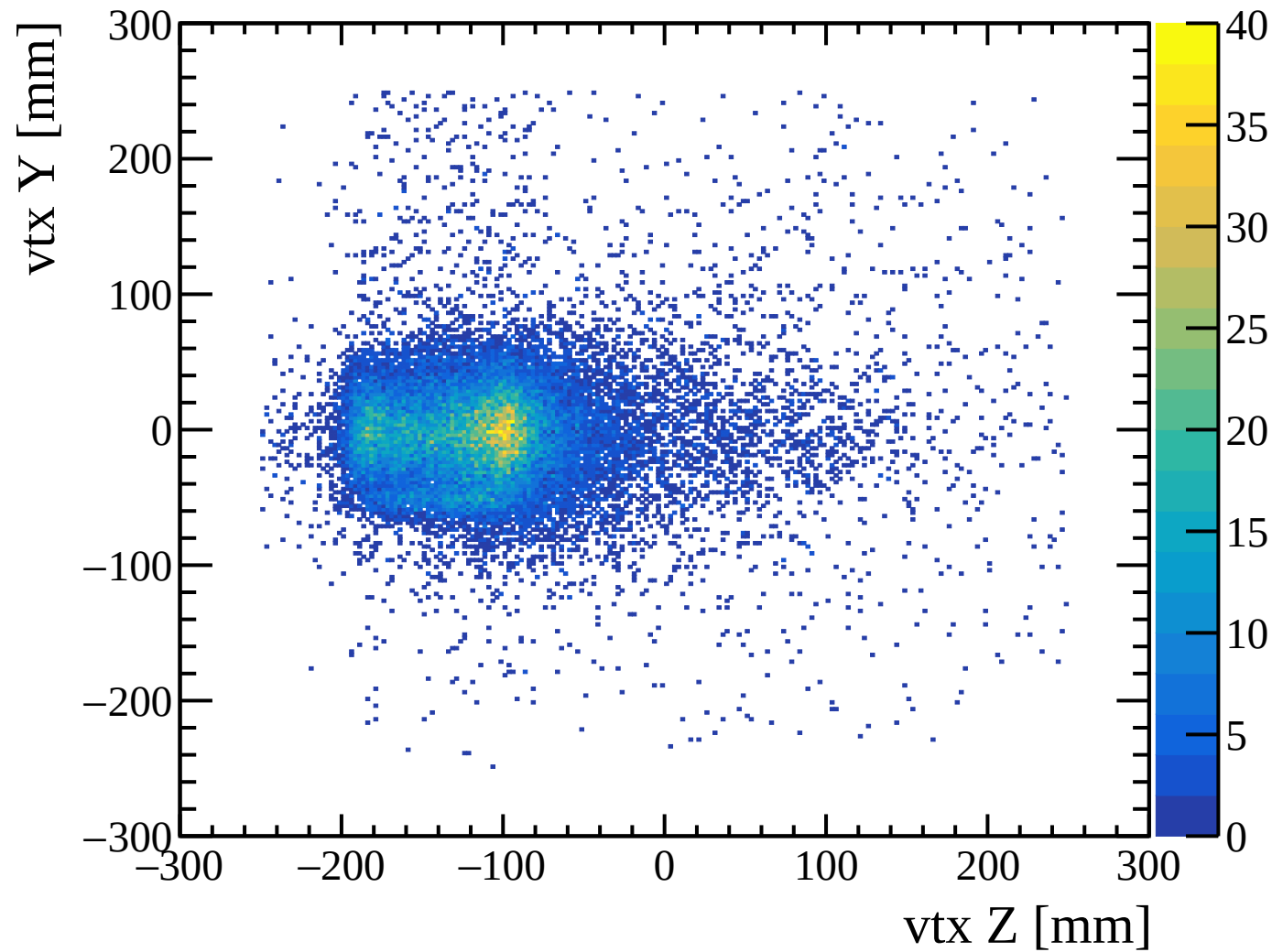
vertex, all  $p\pi^-$



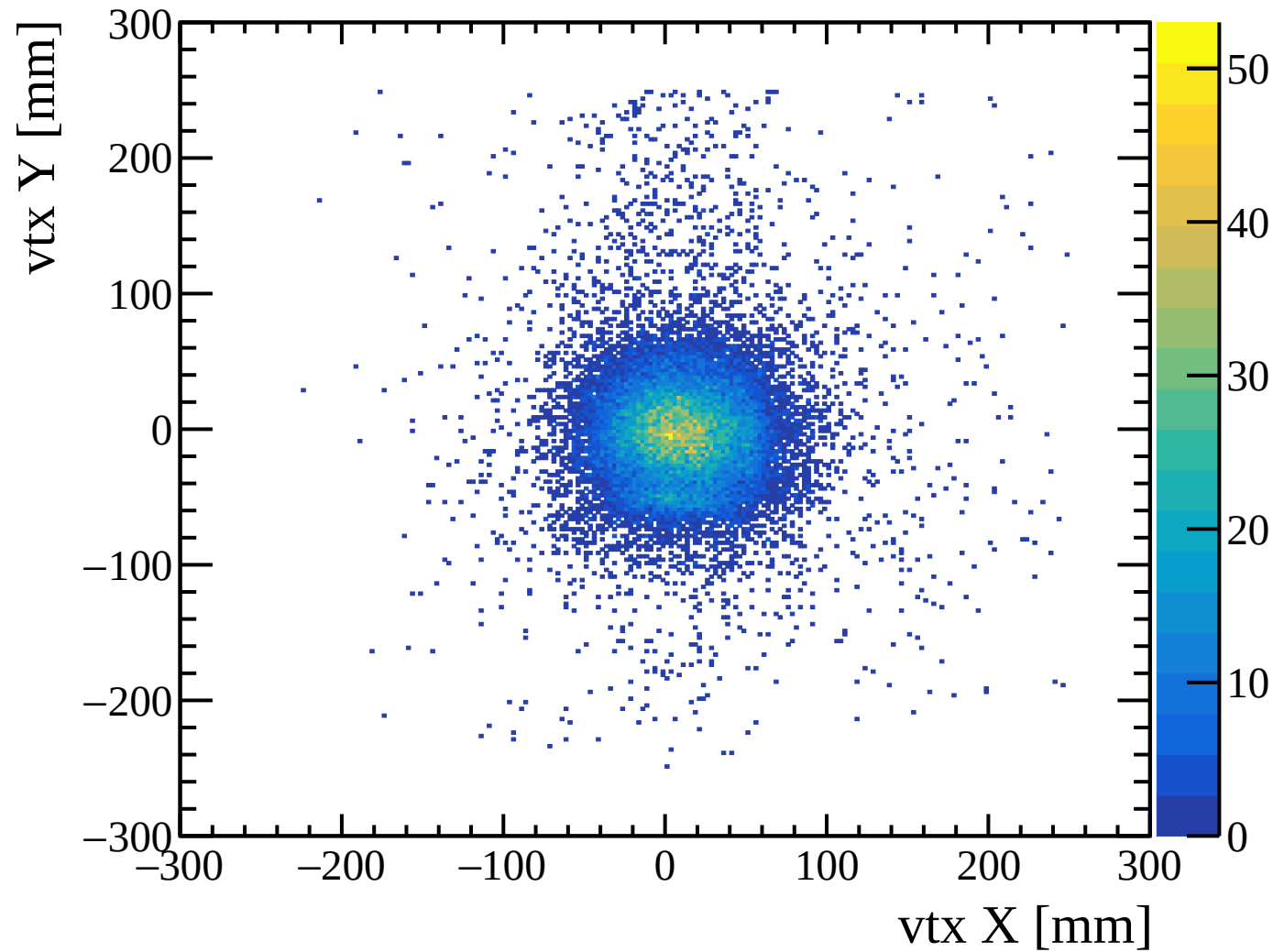
vertex, Lambda peak



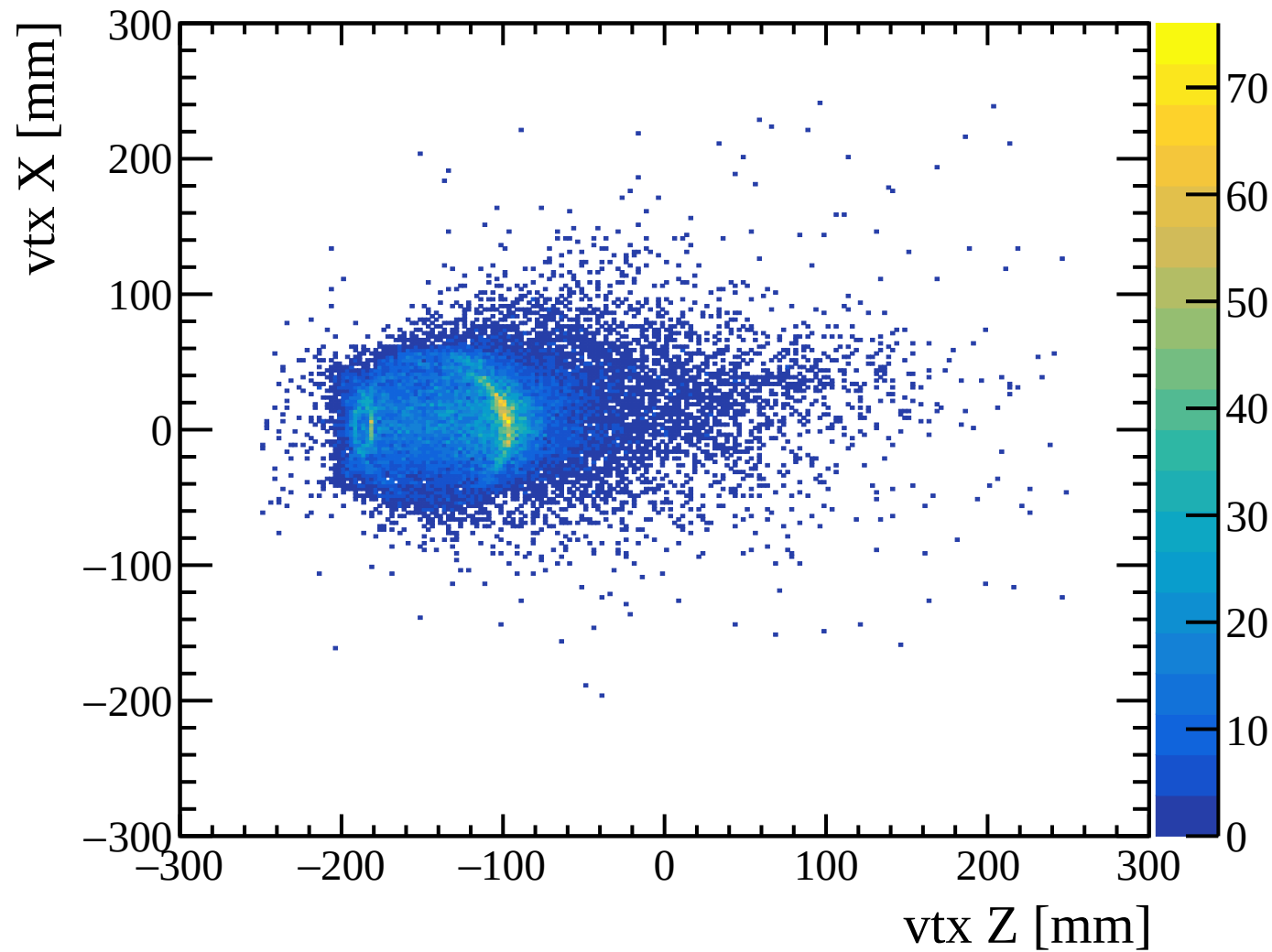
vertex, Lambda peak



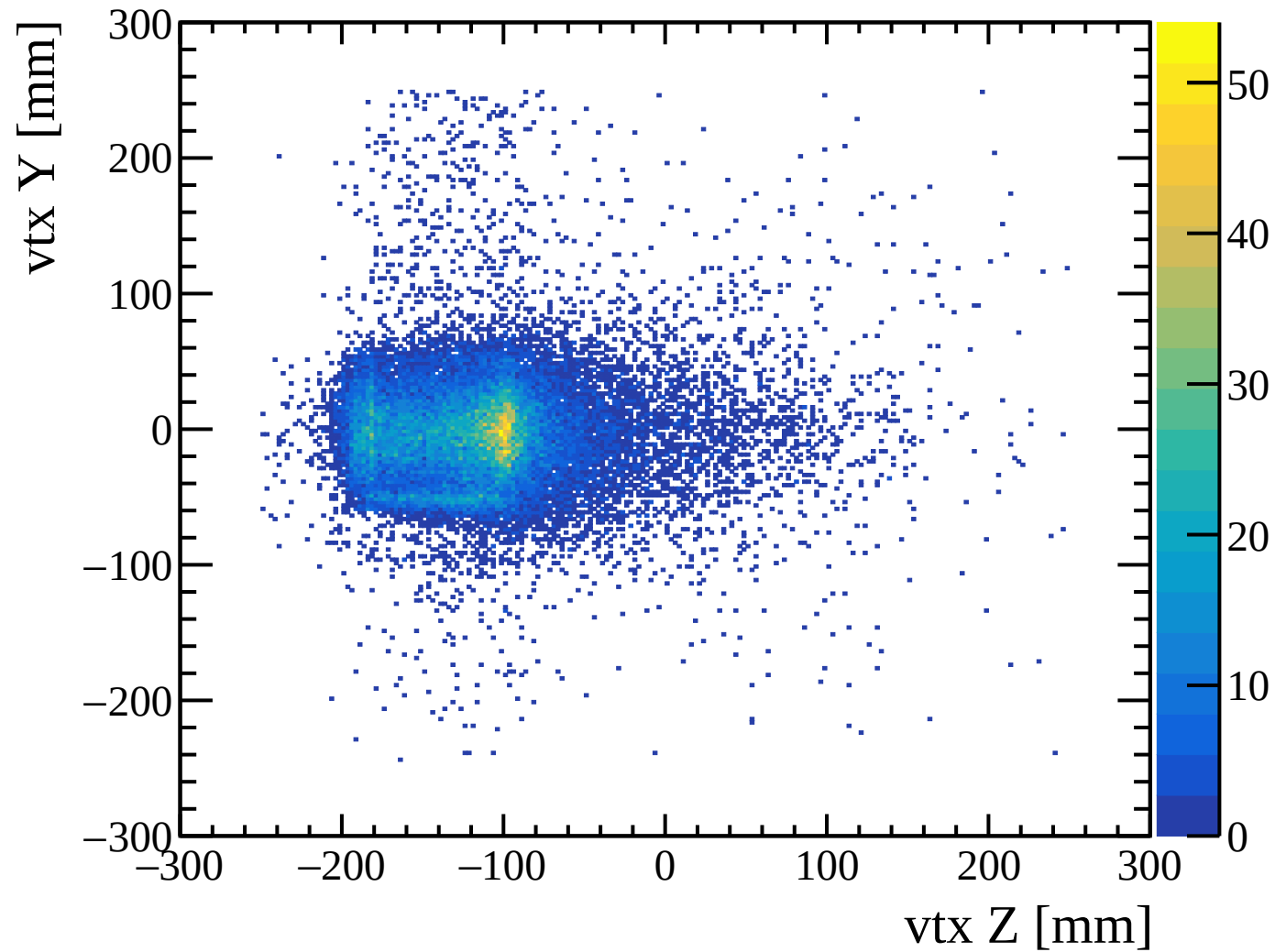
vertex, Lambda peak



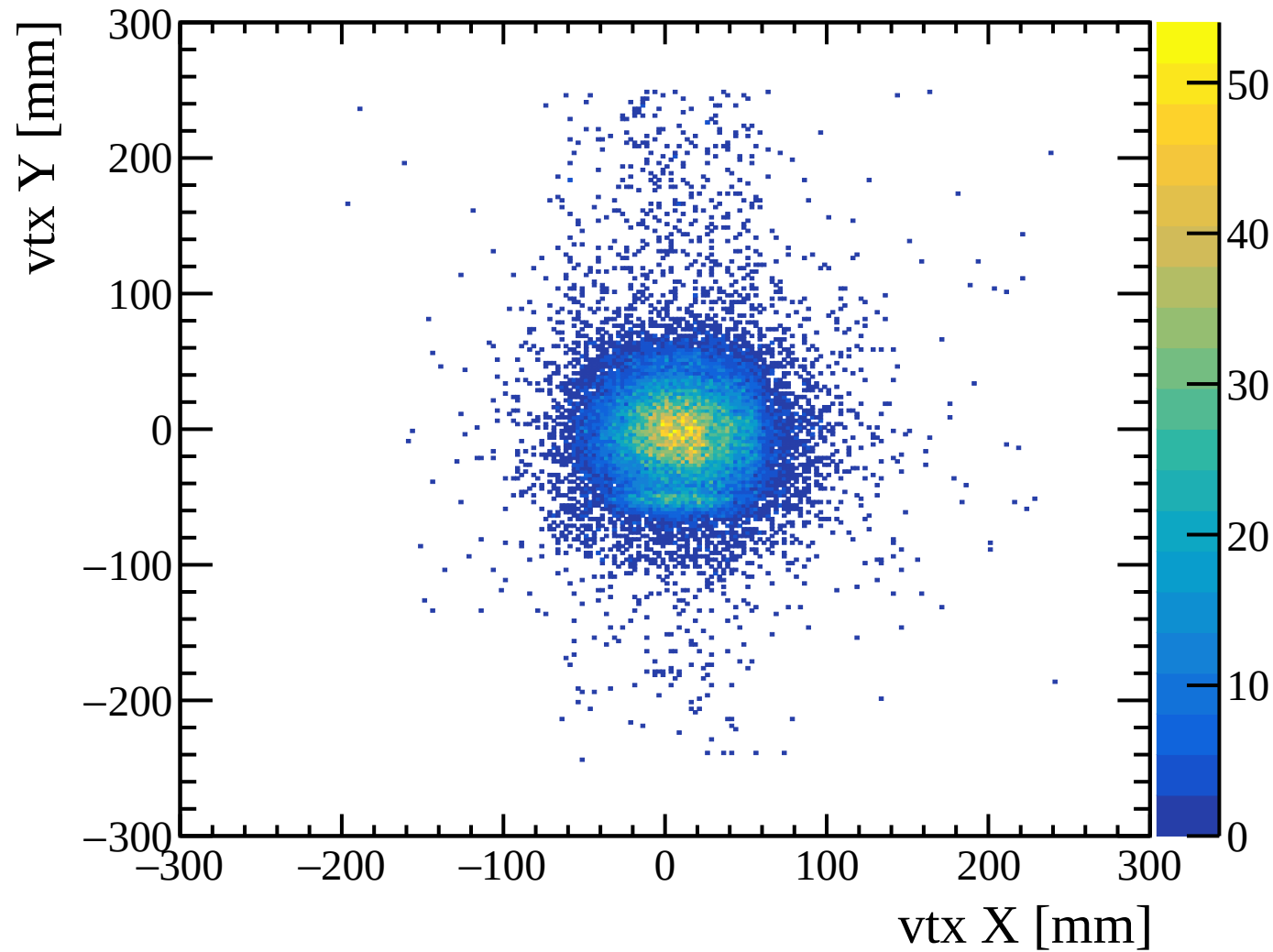
vertex, tight cut



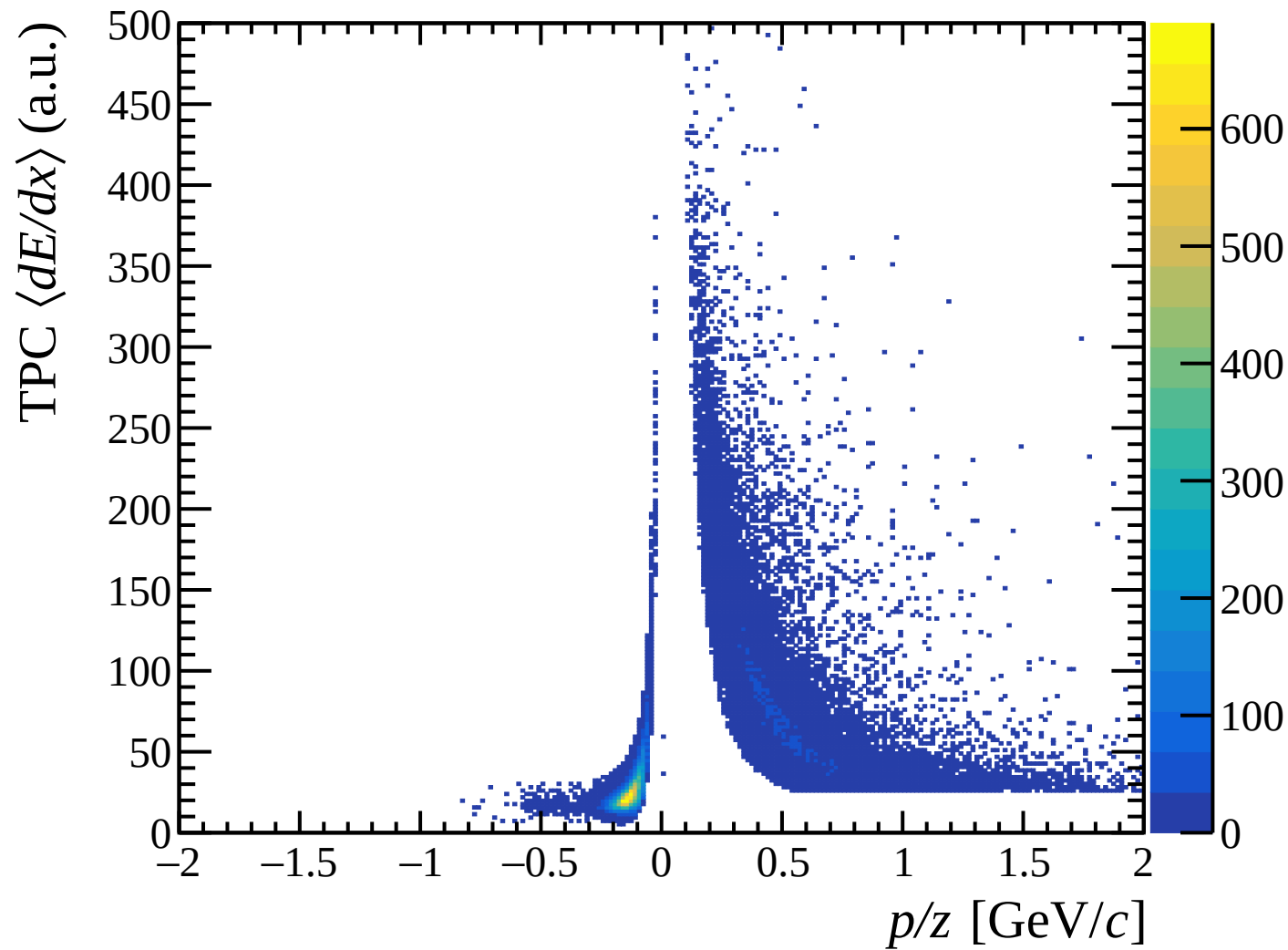
vertex, tight cut



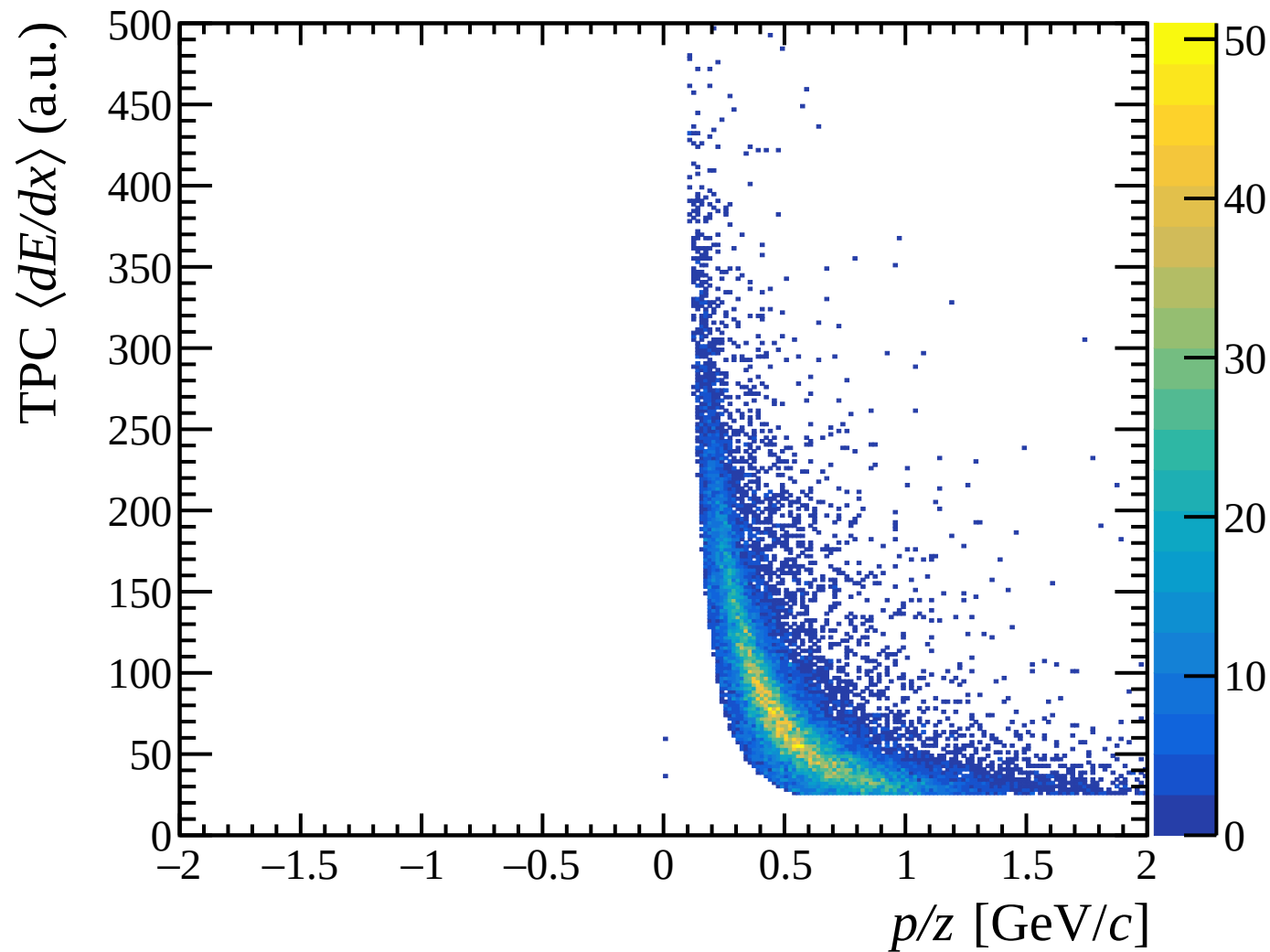
vertex, tight cut



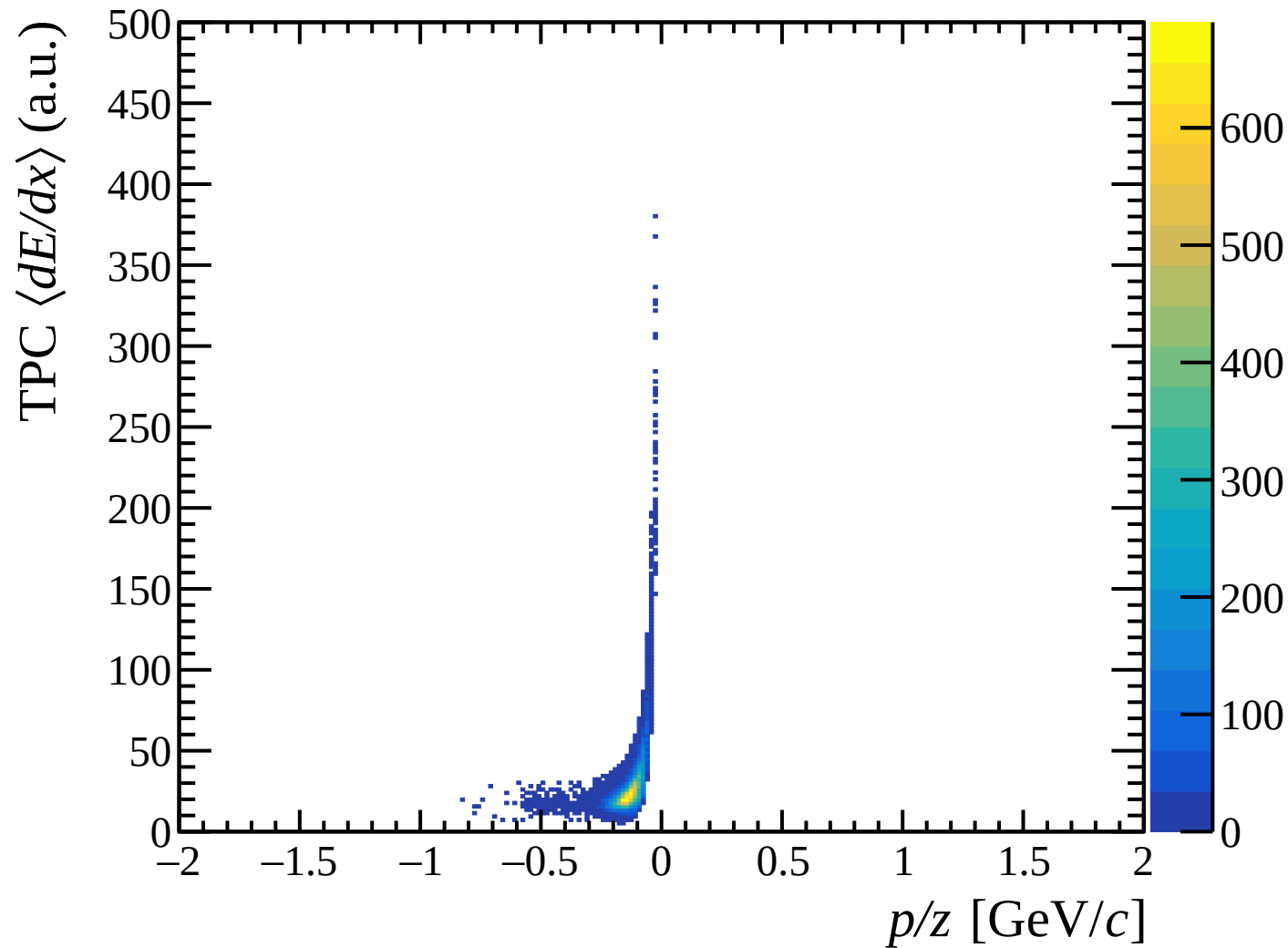
dE/dx vs signed momentum, Lambda peak



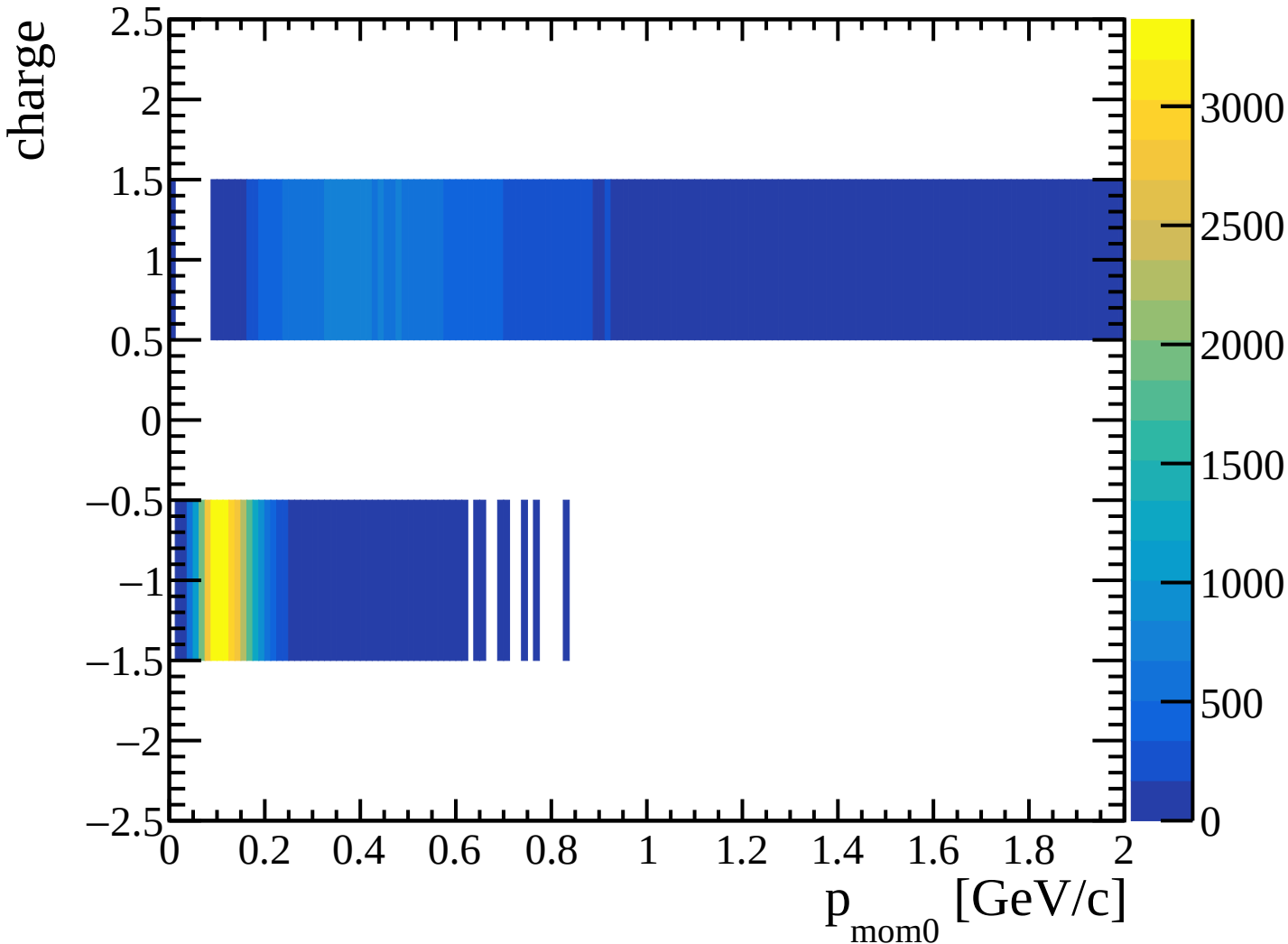
proton  $dE/dx$  vs signed momentum, Lambda peak



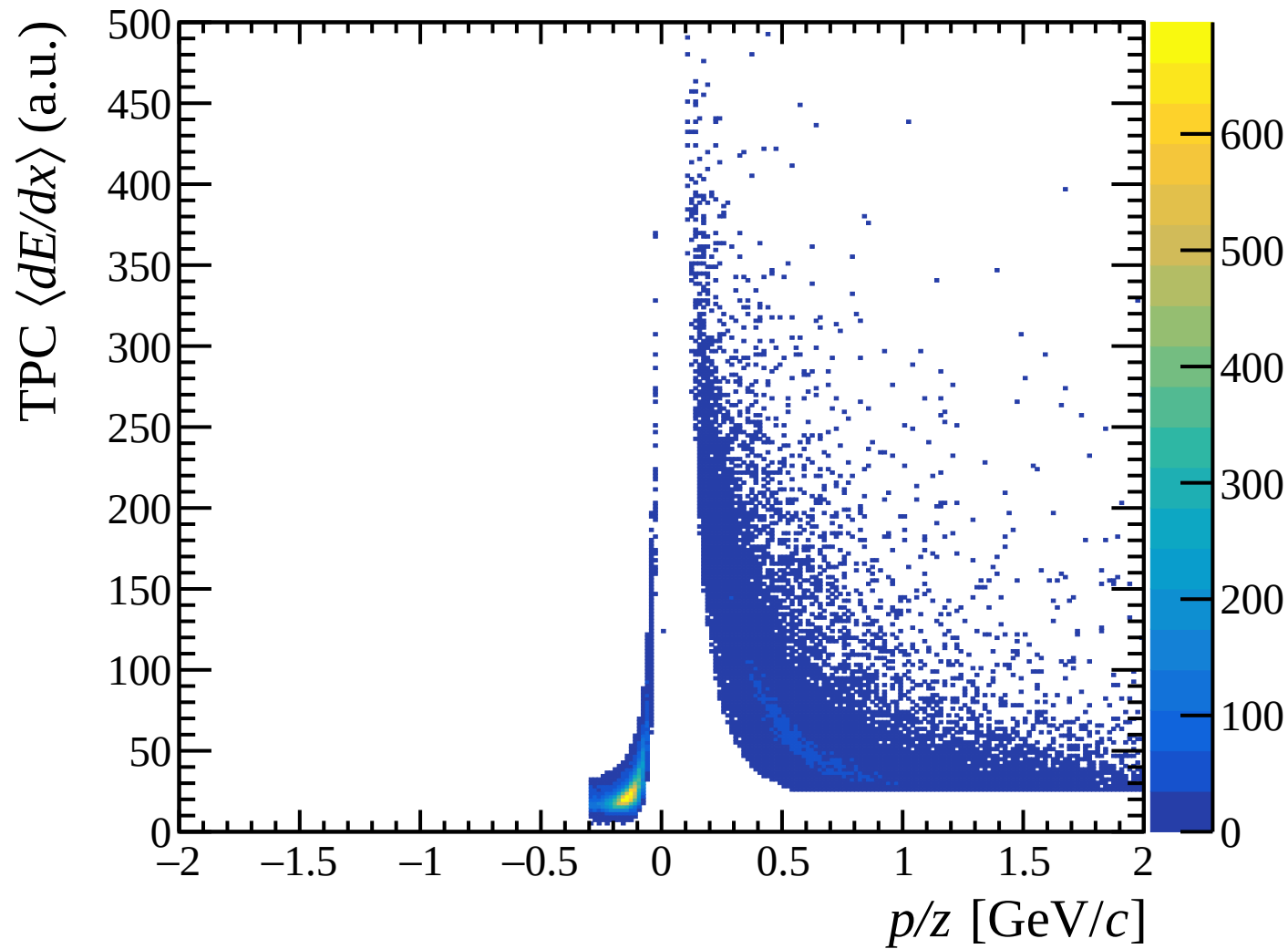
$\pi^-$   $dE/dx$  vs signed momentum, Lambda peak



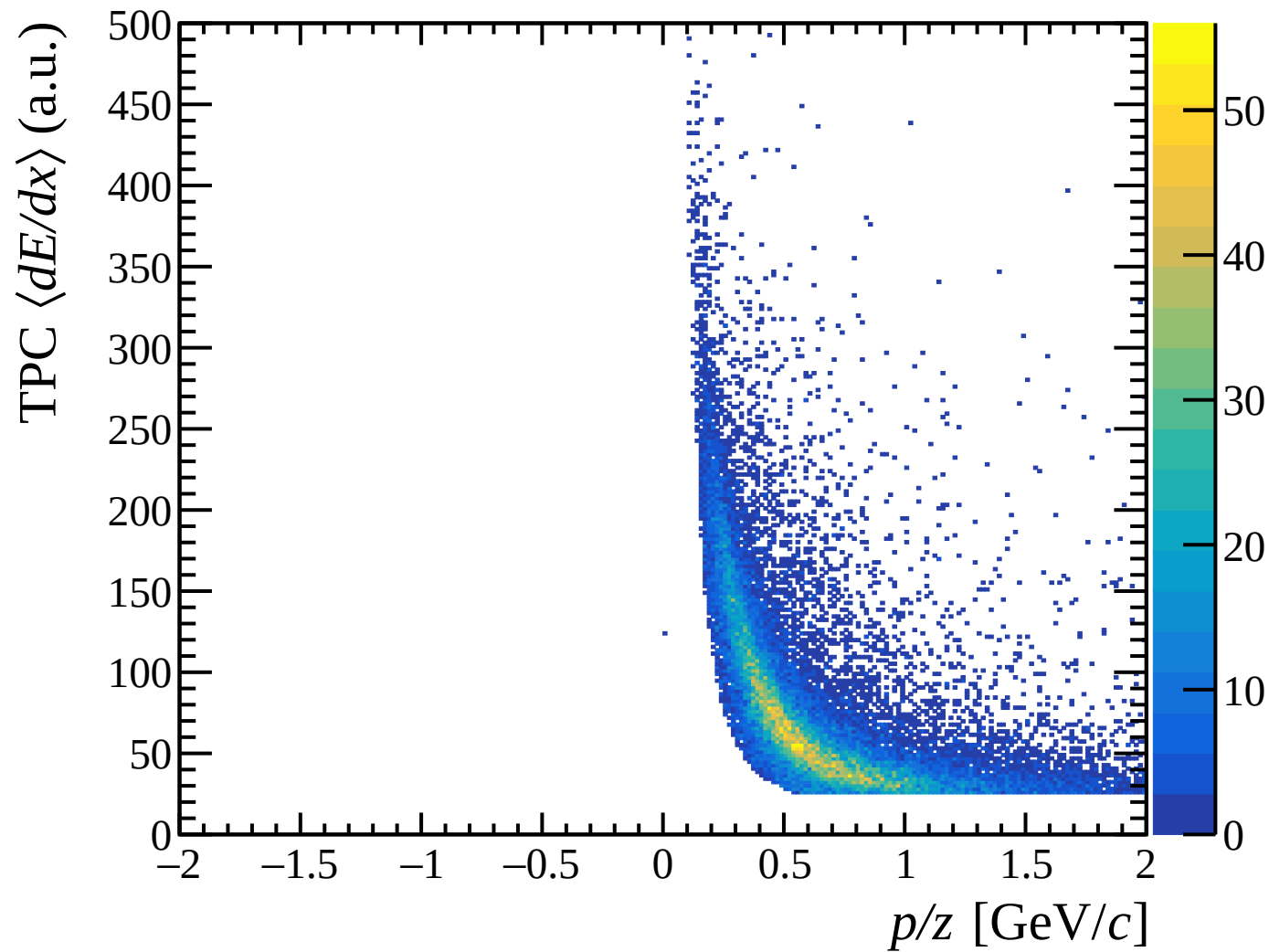
charge vs momentum, Lambda peak



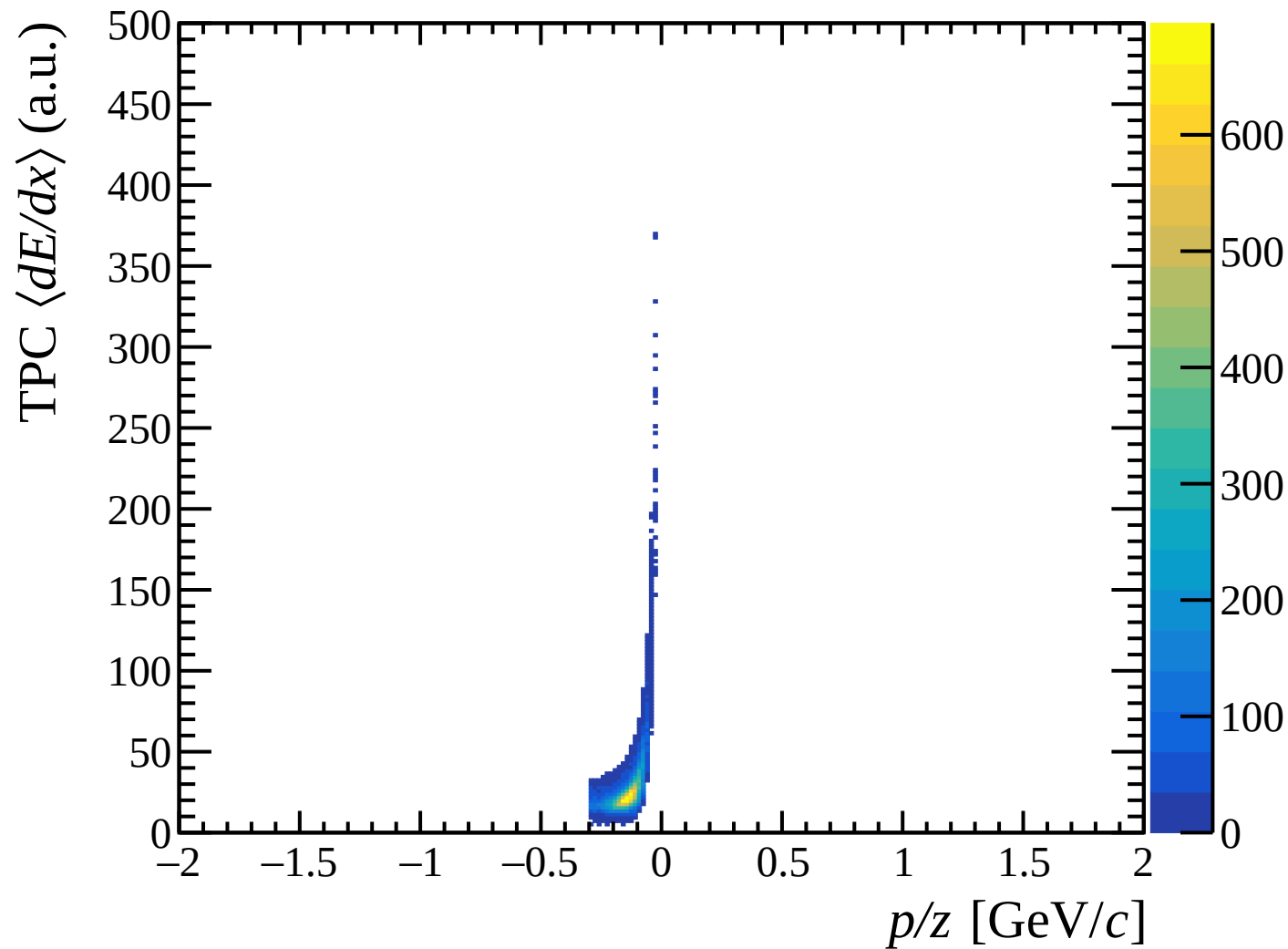
dE/dx vs signed momentum, tight cut



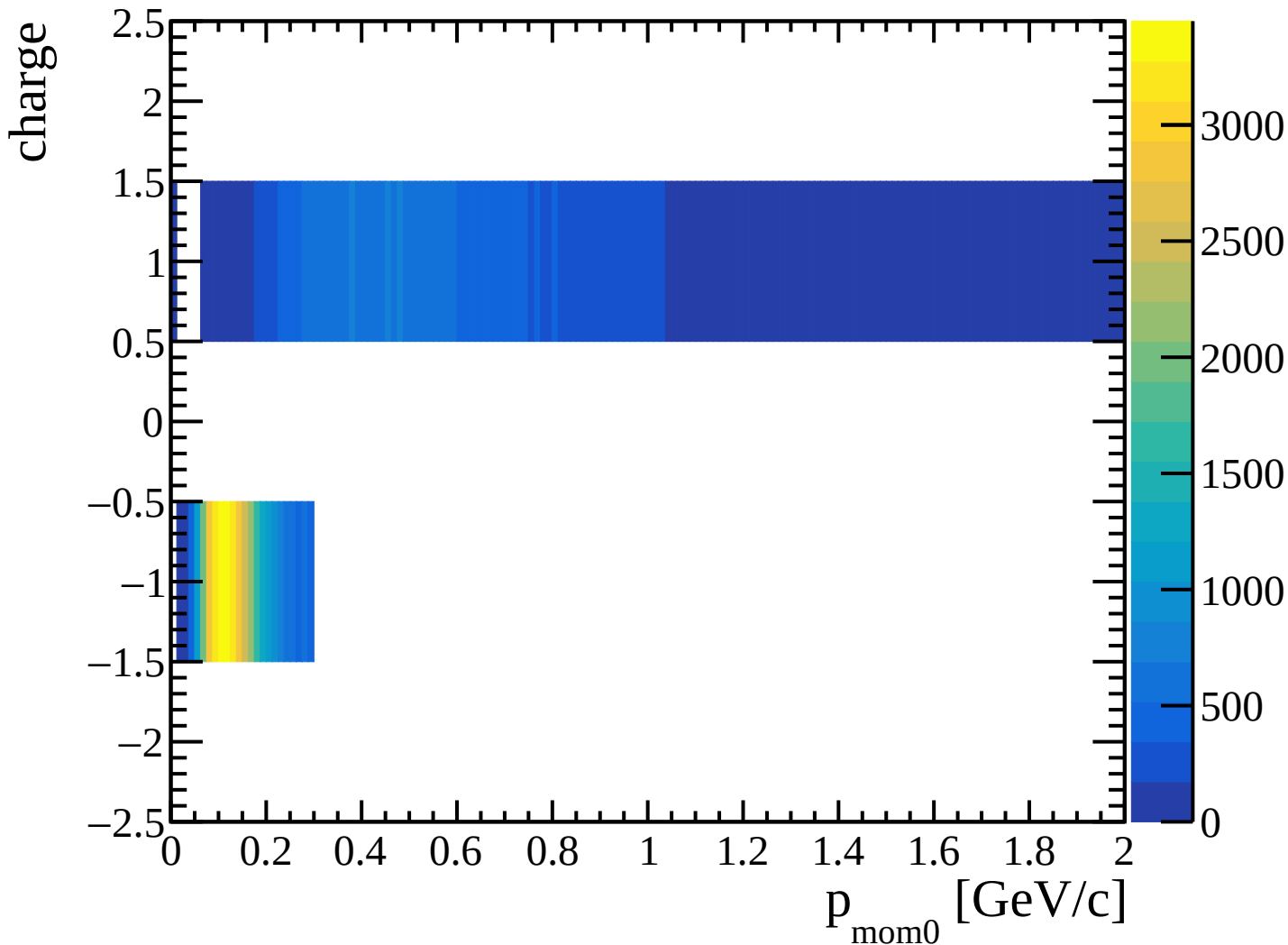
proton  $dE/dx$  vs signed momentum, tight cut



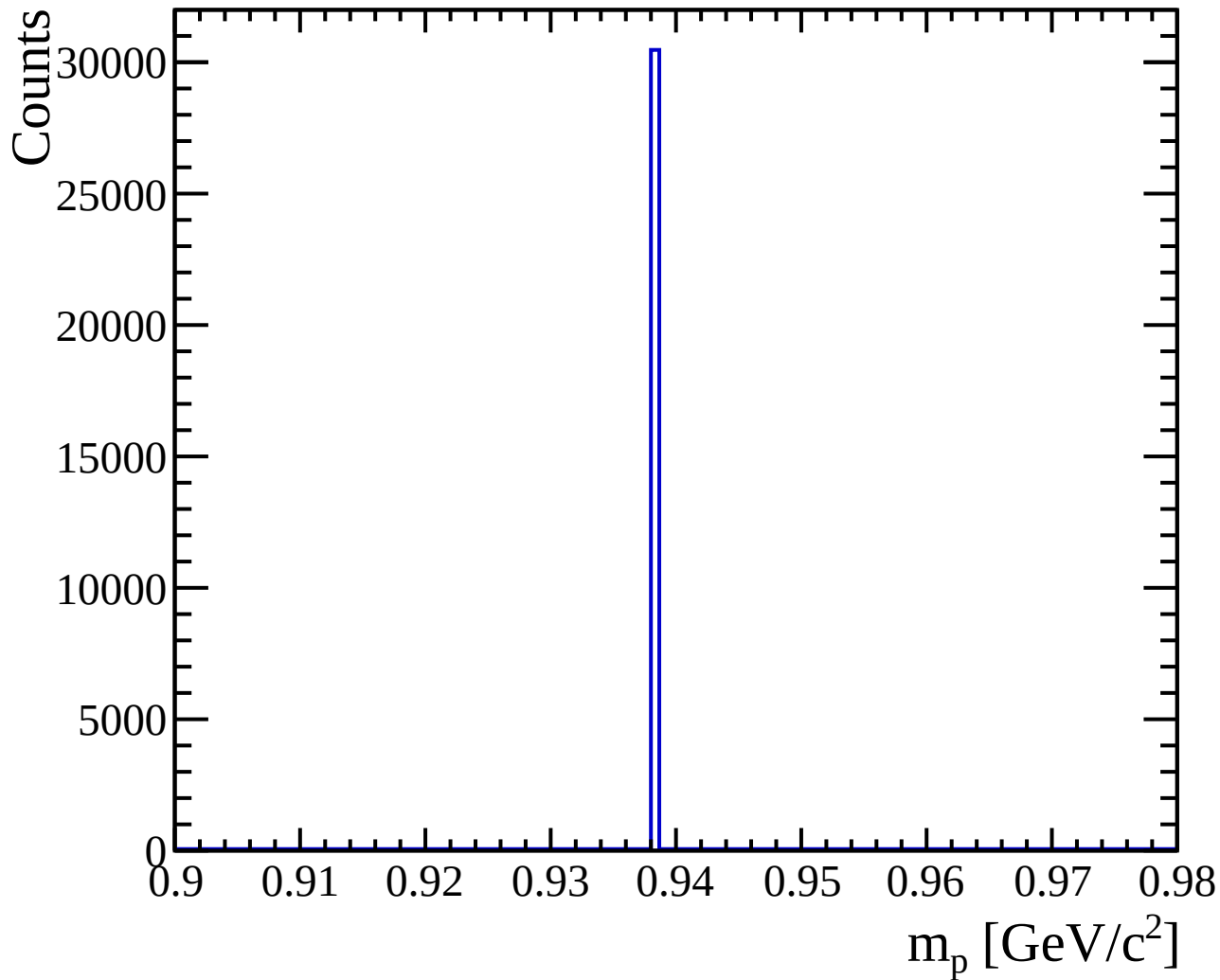
$\pi^-$   $dE/dx$  vs signed momentum, tight cut



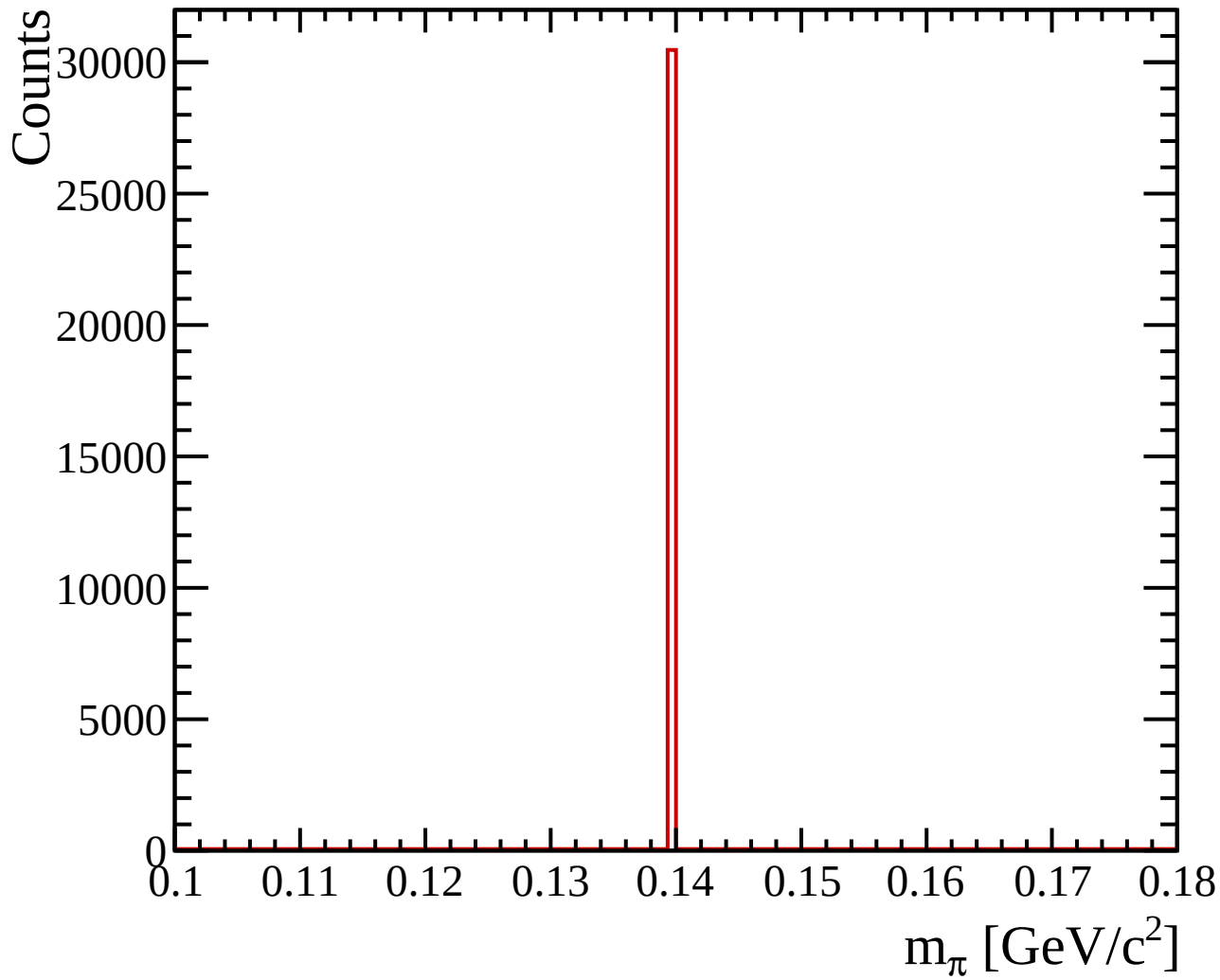
charge vs momentum, tight cut



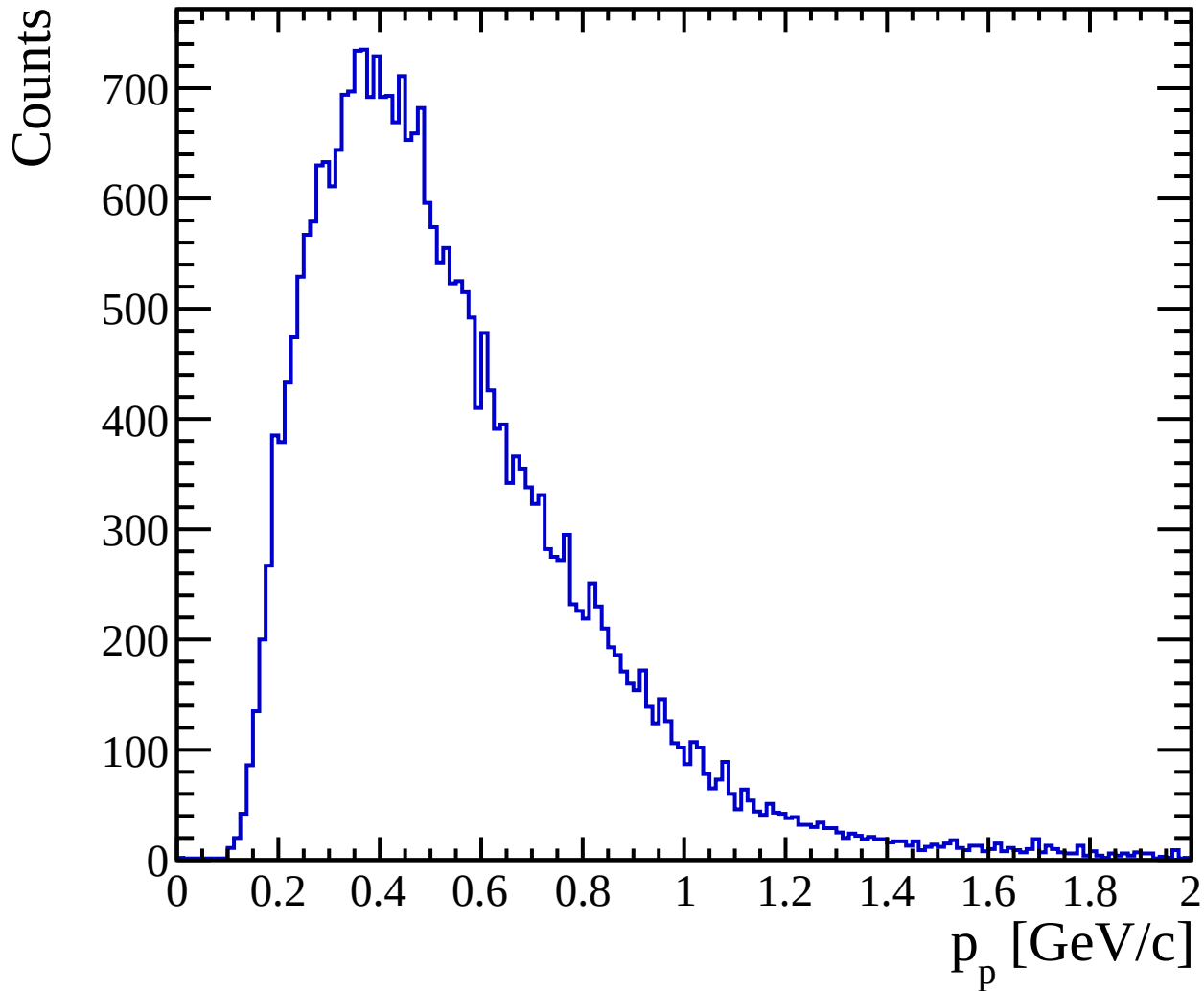
proton candidate invariant mass, Lambda peak



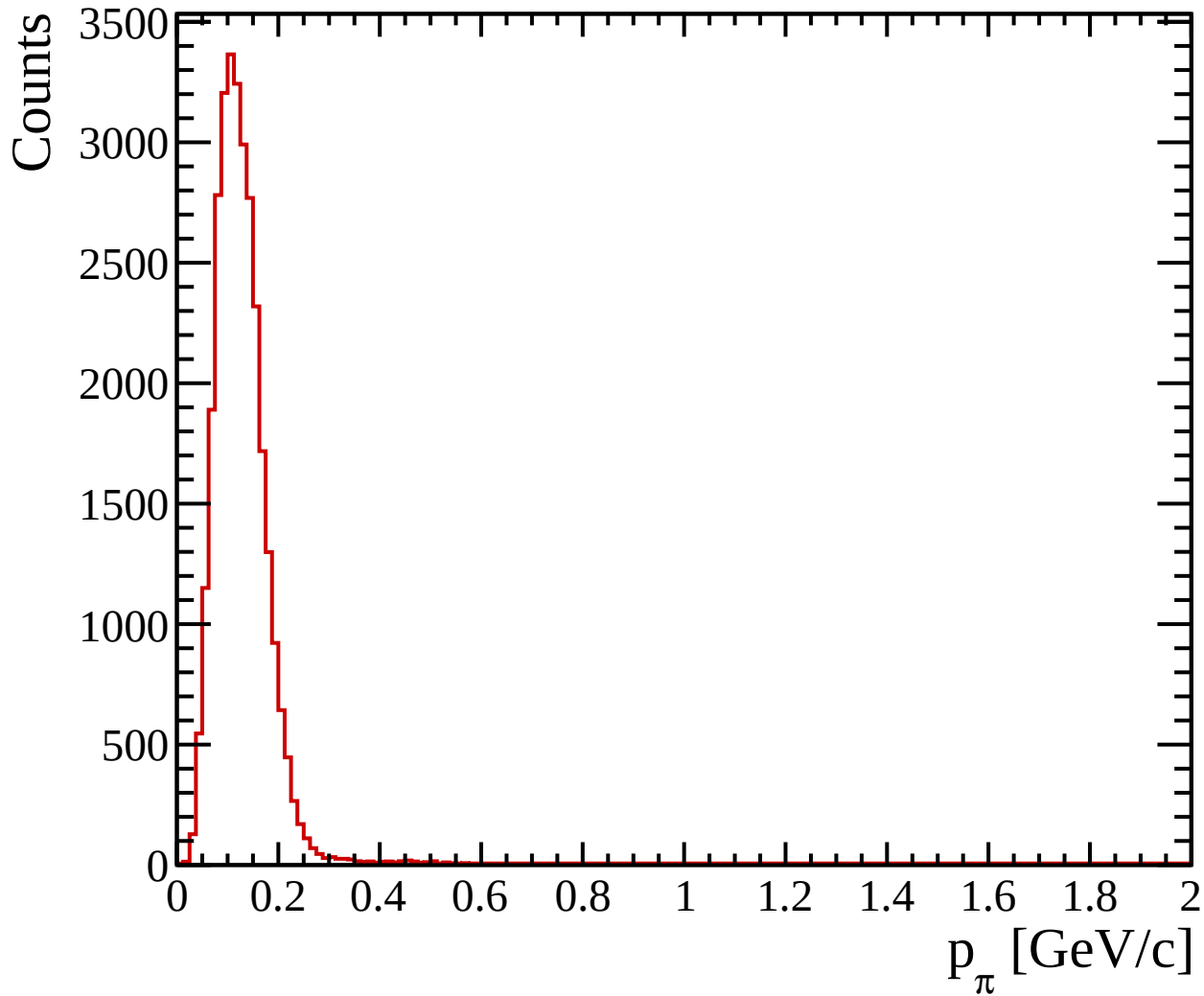
$\pi^-$  candidate invariant mass, Lambda peak



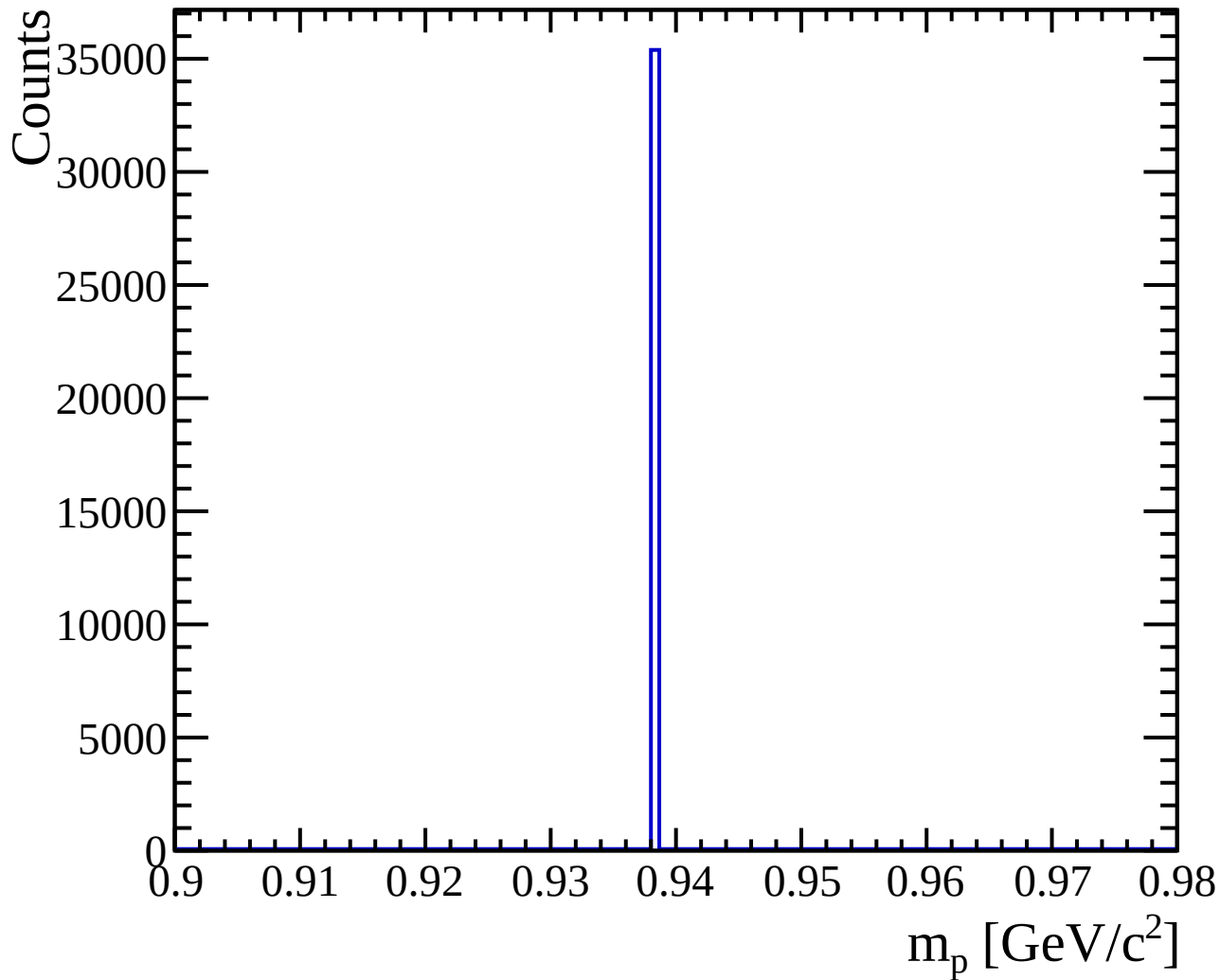
proton momentum at mom0, Lambda peak



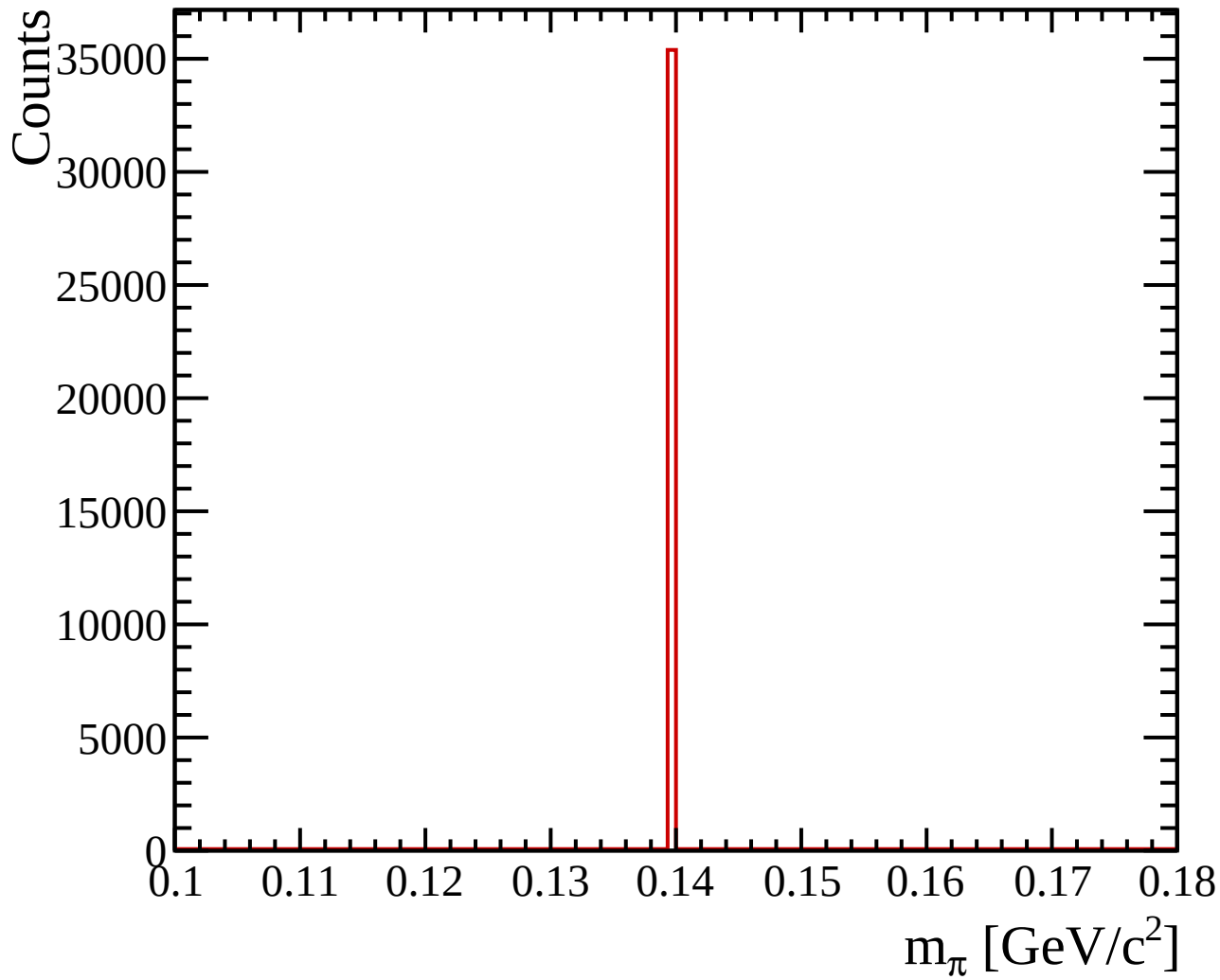
$\pi^-$  momentum at mom0, Lambda peak



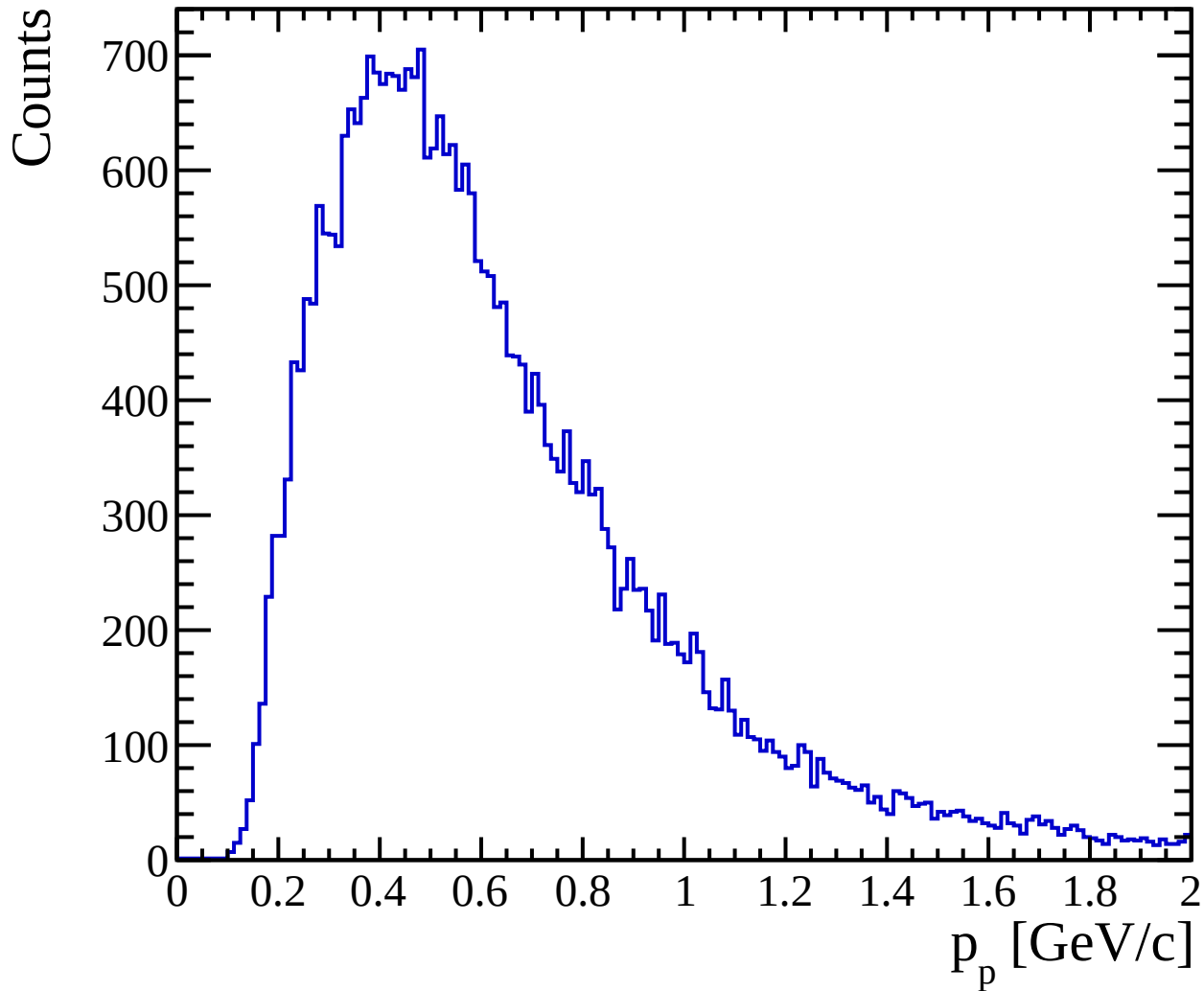
proton candidate invariant mass, tight cut



$\pi^-$  candidate invariant mass, tight cut



proton momentum at mom0, tight cut



$\pi^-$  momentum at mom0, tight cut

